Laws of Motion
&
Friction

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Laws of Motion

Kinematics is the study of motion without considering the causes of that motion, which we discussed in previous chapter.
Now, we are moving to Dynamics which is the study of motion with considering the causes of that motion. It includes study of force, torque and their effects on motion.
The cause for any motion to start or stop is force. Force is defined as a push or pull which can either change or show a tendency to change the state of rest or the state of uniform motion of a body.
Force is a vector quantity. SI unit of force is newton (N) and CGS unit of force is dyne.

$1\ N = 10^5$ dyne

A force can have following effects on the motion of a body:

a. It can change the speed of the moving body.
b. It can change the speed as well as the direction of motion of the body, or
c. It can change only the direction of motion while keeping the speed same, or
d. It can change the shape and size of the body.

Frame of reference

A frame of reference is the location from where the motion of a body is observed. The frames of reference are of two types:

a. Inertial frame of reference: It is a frame which is either at rest or is moving with uniform velocity.

b. Non-inertial frame of reference: It is a frame which is either accelerating or decelerating.
The explanation of motion of an object greatly depends on the position of the observer. So, it is important to identify the frame of reference for studying the motion of a body.
To explain the motion of bodies, Issac Newton gave three laws of motion which examine the effect of forces on bodies.

Newton's Laws of Motion:

(a) **Newton's First law:** As per this law, if a body is in a state of rest or in a state of uniform motion, then it continues to be in that state until an external force acts on it.
In other words, it can be said that if the net external force on a body is zero, then there will be no change in its state of rest or state of uniform motion.
If $(F_{net})_{ext} = 0$, then there is no change in the state of body.

Q. A passenger while deboarding a bus, falls in the direction of the motion of the bus. What is the reason for it?

Sol. As per Newton's first law, the body of the passenger continues to be in its state of motion same as the bus, but his feet suddenly come to rest. As a result, he falls in the direction of motion of the bus. This is called inertia of motion.

Q. **A car driver feels a sudden backward force when he accelerates his car from rest. What could be the possible reason?**

Sol. As per Newton's first law, the body of the driver continues to be in its state of rest while the car is accelerated suddenly. So, he feels a backward force on him and is moved slightly backwards. This is called inertia of rest.

Q. **Mud struck to the wheels of a bicycle tyre flies off tangentially while the bicycle is pedalled on road. What could be the possible reason?**

Sol. As per Newton's first law, the mud on the bicycle wheel keeps its tangential velocity and is thrown off from the tyre. This is called inertia of direction.

(b) **Newton's Second Law:** If net external force on a body is not zero, then body has acceleration in the direction of force. The relation between the force and the acceleration is,

Force, $\vec{F} = m\vec{a}$

Acceleration, $\vec{a} = \frac{\vec{F}}{m}$

A more specific form of Newton's 2nd law is that the force acting on a body is equal to the rate of change of linear momentum of the body.

$$\vec{F} = \frac{d}{dt}(\vec{p}) = \frac{d}{dt}(m\vec{v}) = \vec{v}\frac{dm}{dt} + m\frac{d\vec{v}}{dt}$$

Q. **A body having a mass of 4 kg starts moving at time t = 0 with an initial speed of 2 m/s in the positive x-axis direction. A constant force of F = 80 N acts on it towards negative x-axis. What is the position of the particle after 5s?**

Sol. Given that, mass of the particle, m = 4 kg

F = −80 N (negative sign indicates the opposite direction of force)

$\therefore$ Acceleration, $a = \frac{F}{m} = \frac{-80\,N}{4\,kg} = -20\,m/s^2$

The position of the particle at any time t can be calculated using equation of motion,

$$x = x_0 + ut + \frac{1}{2}at^2$$

The position of the particle at t = 0 is 0, therefore $x_0 = 0$.

$$\therefore \quad x = ut + \frac{1}{2}at^2$$

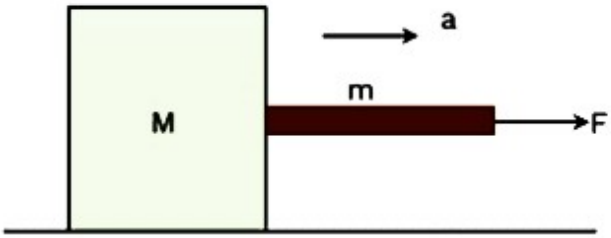
Here, u = 2 m/s, a = −20 m/s_2, t = 5 s,

$$\therefore \quad x = (2 \times 5) + \frac{1}{2}(-20)(5)^2$$

$$= 10 - 250 = -240 \text{ m}$$

Q. **A rope of mass m is used to pull a block of mass M on a horizontal friction-less surface. If a force F is applied at the free end of the rope, then the force exerted by the rope on the block is?**

Sol.



If a is the acceleration of the rope and the block, then

$$a = \frac{F}{(M+m)}$$

The force exerted by the rope on the block is then calculated as,

$$F' = Ma = \frac{MF}{(M+m)}$$

Q. **A conveyer belt is moving with uniform speed of 2 m s^{-1}. Sand is being poured on the conveyor belt at a rate of 5 kg s^{-1}. What is the constant external force needed to maintain the constant speed of the belt.**

Sol. As per Newton's 2nd law,

$$F = m\left(\frac{dv}{dt}\right) + v\left(\frac{dm}{dt}\right)$$

As the speed of the conveyor belt is constant, so $\left(\frac{dv}{dt}\right) = 0$

Then, force required, $F = v\left(\frac{dm}{dt}\right) = 2 \times 5 = 10N$

Q. **A force F given by the empirical law $F = \frac{R}{t^2}v$, where R is constant, v is velocity and t is time, acts on a particle of mass m. Find a relation between velocity of the particle and time.**

Sol. Given that, force, $F = \frac{R}{t^2}v$

By Newton's 2nd law, $F = m\left(\frac{dv}{dt}\right)$

So, $m\frac{dv}{dt} = \frac{R}{t^2}v \Rightarrow \frac{dv}{v} = \frac{R}{m}\frac{dt}{t^2}$

Integrating both sides,

$$\int\frac{dv}{v} = \frac{R}{m}\int\frac{dt}{t^2} \Rightarrow \ln v = -\left(\frac{R}{m}\right)\left(\frac{1}{t}\right) + C$$

Q. **A block having a mass of 15 kg is acted upon by two forces each having magnitude of 10 N making an angle of 60° with each other. Find the magnitude of net acceleration of the body.**

Sol. As force is a vector quantity, so resultant force acting on the given block will be, $F_{resultant} = \sqrt{(F)^2 + (F)^2 + 2F^2\cos 60^\circ}$

$$= \sqrt{3F^2} = \sqrt{3}F$$

By 2nd law, $ma = \sqrt{3}F$ or, Acceleration, $a = \frac{\sqrt{3}F}{m}$

Here, F = 10 N, m = 15 kg,

$$\therefore \quad a = \frac{\sqrt{3}\times 10}{15} = \frac{2}{\sqrt{3}}\ m/s^2$$

(c) **Newton's Third law:** As per this law, for every action force, there is an equal and opposite reaction force, the two forces being on different bodies.

$$\vec{F}_{12} = -\vec{F}_{21}$$

Important points to note about Newton's 3rd Law:

(i) The action and reaction forces must act on two different bodies.

(ii) The action and reaction forces should be equal and opposite.

(iii) The line of action of the action and reaction forces should be same.

(iv) The action and reaction forces must be of same nature.

Nature of Forces:

Depending upon nature, there are 4 types of forces

(A) Weak forces

(B) Nuclear forces

(C) Gravitational forces

(D) Electromagnetic forces

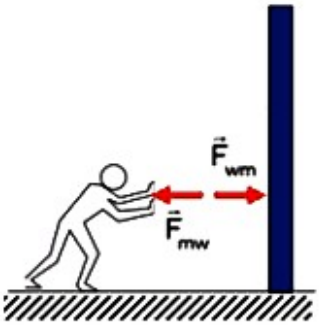
Note : Forces like normal reaction, tension in string (or rope), frictional force, spring force etc fall under the category of "Electromagnetic forces".

Explanation of Newton's Third Law: Consider a man is pushing a fixed wall as shown in figure below.



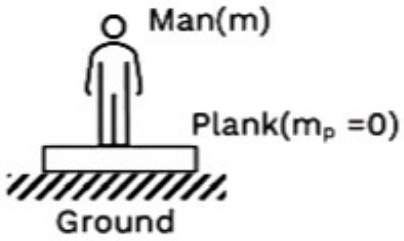
If $\vec{F}_{wm}$ is the force on wall applied by the man and $\vec{F}_{mw}$ is the force on man by the wall, then by 3rd law of motion,

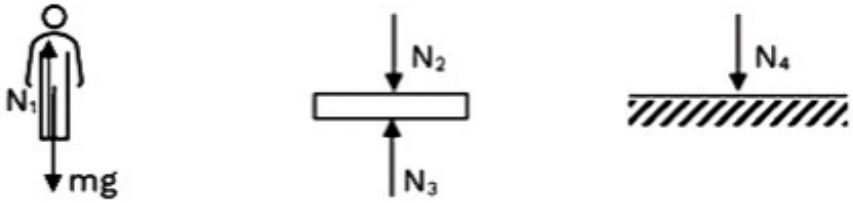
$\vec{F}_{wm} = -\vec{F}_{mw}$



$\vec{F}_{mw}$ and $\vec{F}_{wm}$ are action reaction pair as these forces are acting on two different bodies, are equal and opposite, have same line of action and are of same nature.
Note that action and reaction pairs can never cancel each other as they act on different bodies.

Now, consider a plank of almost zero mass is kept on ground and a man of mass m is standing on the plank as shown.
Let's draw free body diagram (F.B.D.) of man, plank and ground.





Note: Free body diagram: A diagram showing all forces acting on the body by removing all supports and replacing the supports by normal reactions.

In the free body diagram of the man and the plank,

N_1 = Normal reaction on the man by the plank

mg = Weight of the man

N_2 = Normal reaction on the plank by the man

N_3 = Normal reaction on the plank by ground

N_4 = Normal reaction on ground by the plank

For the man, $N_1 = mg$ ...($\because$ Man is standing at rest, so $\vec{a}_{man} = 0$)

$$\left\{\begin{array}{l} \therefore \vec{F}_{net} = 0 \\ N_1 - mg = 0 \\ N_1 = mg \end{array}\right\}$$

Similarly, for the plank, $N_2 = N_3$ ....($\because$ $\vec{a}_{plank} = 0$)

Also, $N_1 = N_2$ ... (By Newton's 3rd law)

and, $N_3 = N_4$ ... (By 3rd Law)

Thus, $mg = N_1 = N_2 = N_3 = N_4$

Q. **Are N_1 and mg action-reaction pair?**

Sol. No, N_1 and mg are not action-reaction pair as they are acting on the same body i.e., the man.

Q. **Are N_1 and N_2 action-reaction pair?**

Sol. Yes, N_1 and N_2 are action-reaction pair because
(i) They are equal and opposite
(ii) Acting on different bodies
(iii) Have same line of action
(iv) Have same nature

Q. **Are N_2 and N_3 action-reaction pair?**

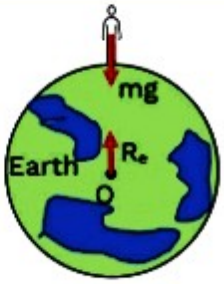
Sol. No, N_2 and N_3 are acting on same body.

Q. **Are N_3 and N_4 action reaction pair?**

Sol. Yes, N_3 and N_4 are action reaction pair. Reason is same as (b).

Q. **Where is the reaction of "mg"?**

Sol. Reaction of "mg" is at centre of earth as this "mg" force arises due to gravitational attraction of earth. As "mg" is gravitational force, so its reaction must be having same nature.
R_e = mg → Both are of same nature.

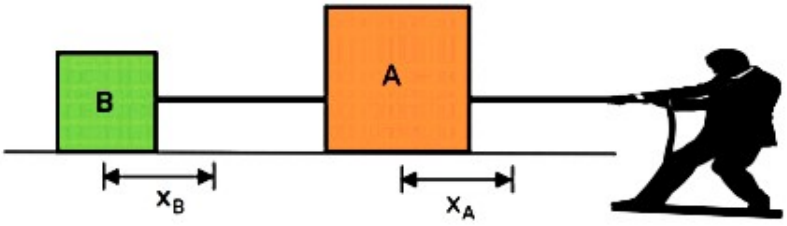


Note:

* Newton's 3rd Law is always true and applicable whether the bodies are at rest or in motion. It is applicable everywhere.
* Laws in physics are based on experimental analysis and are found to be true everywhere. It is incorrect to ask for any mathematical proof for a law in Physics..

Constraint Motion

Constraint motion means "bound" motion.
Let's say we have two blocks kept on ground connected by a rope as shown

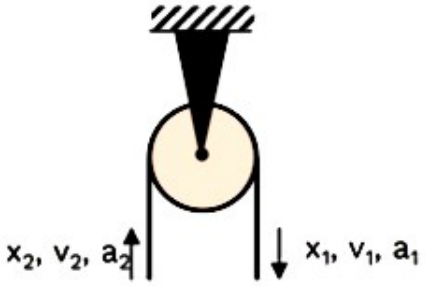


Now, if a man pulls the block A by a distance x_A, then due to the connection of rope, B also has to move by $x_B = x_A$. Thus, we can say that B is bound to move if A is moved. Such a motion is known as constraint motion.

Two or more bodies can be set in a constraint motion in several ways. One of the many methods is to use pulleys.

Constraint relations for pulleys:

a) Fixed pulley:



$$x_1 = x_2$$
$$v_1 = v_2$$
$$a_1 = a_2$$

For the fixed pulley shown in above figure, if the string on one side moves by a distance x_1 in downward direction, then the other end must move by a same distance $x_2 = x_1$ as the string is considered to be inextensible. Same relation follows for velocity and acceleration as well.

b) For a movable pulley

For a movable pulley, we can have three moving ends as shown in the figure below. To find the relation between the displacements of the three points, we can do a simple displacement of the pulley as shown. Displace the pulley upwards by a distance x_A. Then the end A moves up by x_A as well.

If B and C are not moving independently, then B and C will also move up by x_A.

From the figure, we can write, $x_B + x_C = 2x_A$

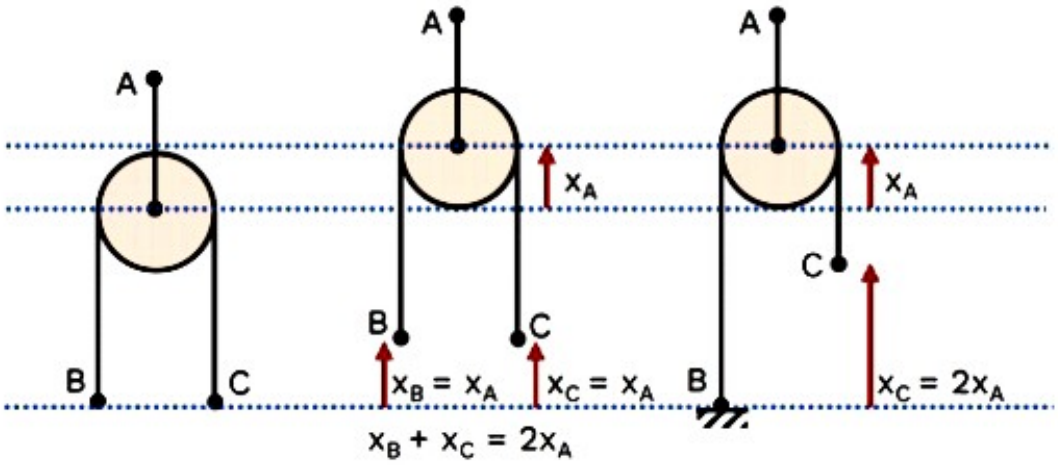
This gives us a constraint relation between the displacements of the three ends A, B and C as,

$$x_A = \frac{x_B + x_C}{2}$$

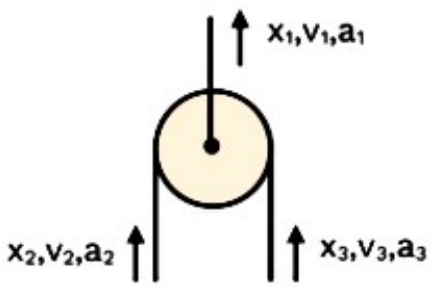
Similar relation follows for velocity and acceleration also.

In the third figure, if the end B is kept fixed, then its displacement will be zero, then in that case, we have,

$$x_A = \frac{0 + x_C}{2} = \frac{x_C}{2}$$



The constraint relation for a movable pulley can be summarised as the following figure,



$$x_1 = \frac{x_2 + x_3}{2}$$

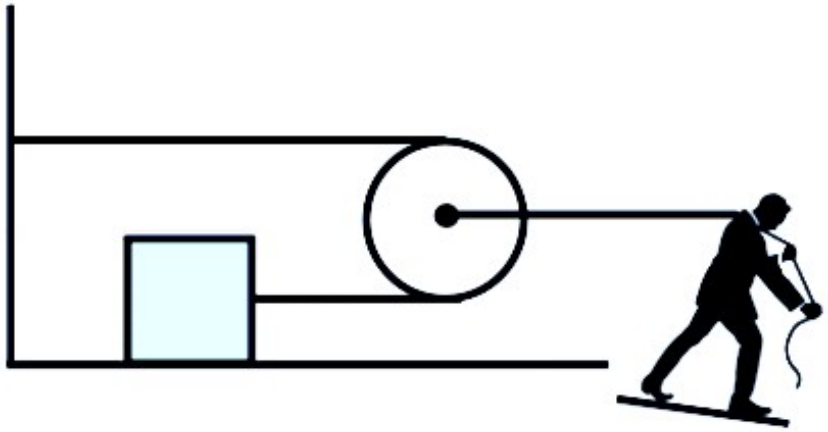
$$v_1 = \frac{v_2 + v_3}{2}$$

$$a_1 = \frac{a_2 + a_3}{2}$$

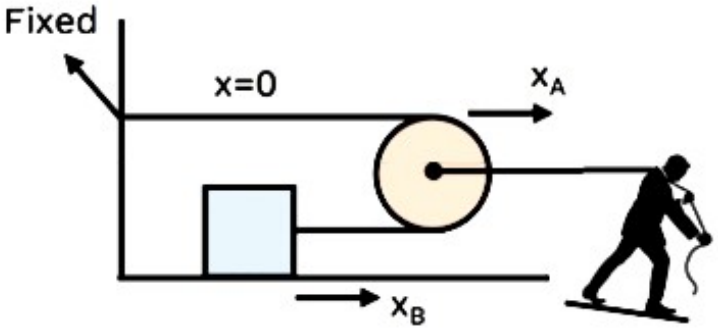
If any direction is opposite, put that value with –ve sign in above formulae.

EXAMPLES ON PULLEY CONSTRAINT RELATIONS

Q. **Find the displacement of the block if man displaces the pulley by 1m.**



Sol.



By constraint relation for movable pulley,

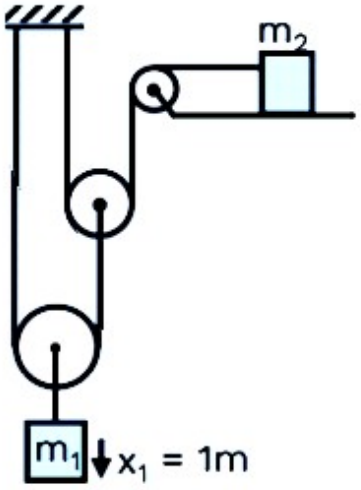
$$x_A = \frac{0 + x_B}{2}$$

$$x_B = 2x_A$$

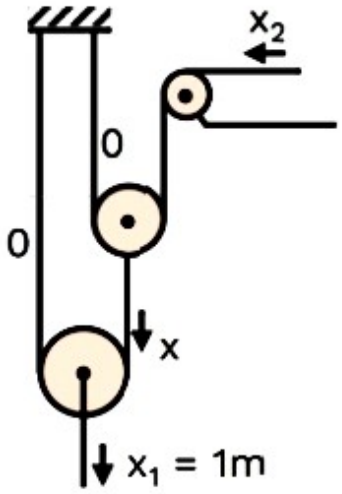
$$= 2\ (1\ m)$$

$$x_B = 2\,m$$

Q. **Find displacement of block of mass m_2 in the given system.**



Sol. By constraint relation for lower pulley,



$$x_1 = \frac{0 + x}{2}$$

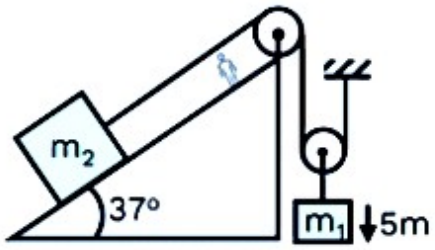
$$x = 2x_1$$

and, by constraint relation for the central pulley, $x = \frac{0 + x_2}{2}$

So, $x_2 = 2x = 2(2x_1) = 4x_1$

$x_2 = 4(1m) = 4m \rightarrow$ Towards *left*

Q. **Find displacement of m2 w.r.t. the man resting on the inclined plane in given system.**



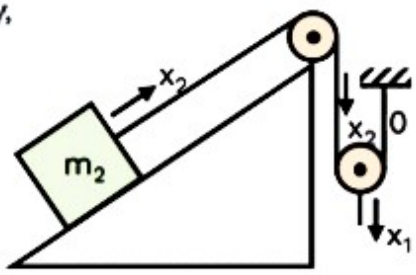
Sol. By constraint relation for the lower movable pulley,

$$x_1 = \frac{0 + x_2}{2}$$

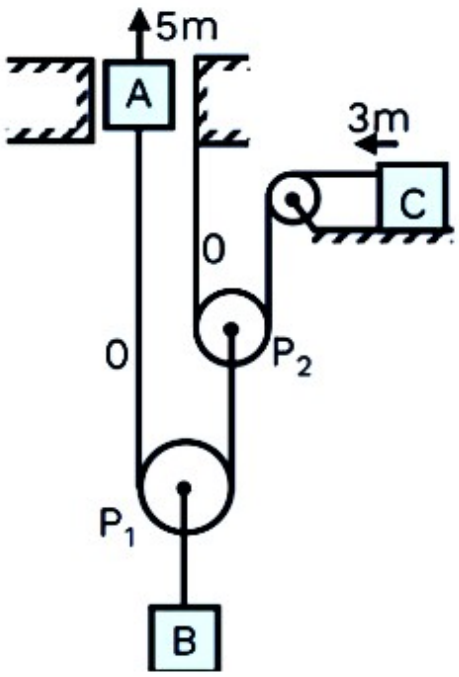
$$x_2 = 2x_1$$

$$= 2(5m)$$

$x_2 = 10m$ (up the inclined plane)



Q. **Find displacement of block B in the mass pulley system given below.**



Sol. For pulley P_2,

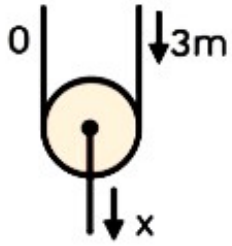
$$x = \frac{0+3}{2} = \frac{3}{2}m$$

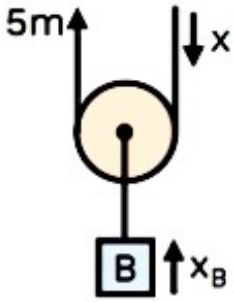
For pulley P_1,

$$-x_B = \frac{x-5}{2}$$

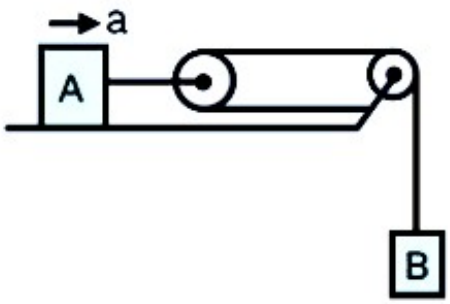
$$-2x_B = \frac{3}{2} - 5$$

$$x_B = \frac{7}{4}m \text{ (upwards)}$$

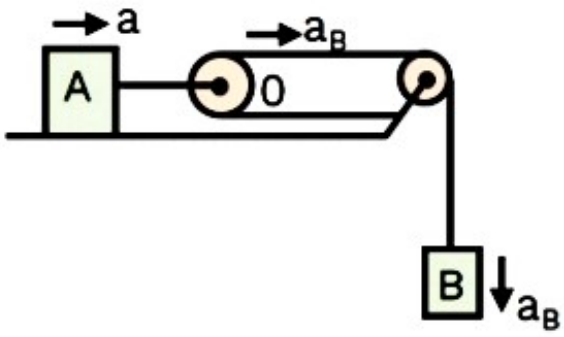




Q. **Find acceleration of block B in the figure given below.**



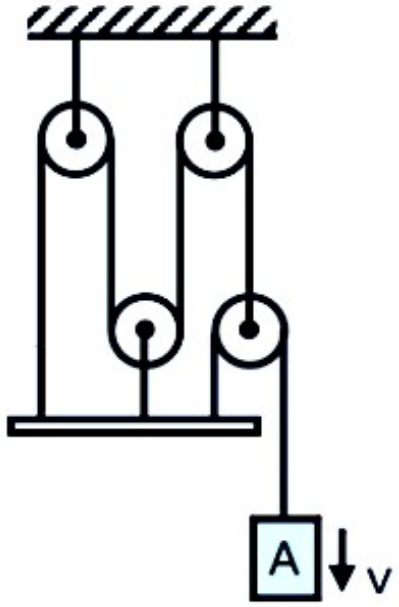
Sol. By constraint relation for the upper movable pulley,



$$a = \frac{0 + a_B}{2}$$

$a_B = 2a \rightarrow$ Downwards

Q. **In given system, the plank always remains horizontal. If the block is moving downwards with velocity v, then find the velocity of plank.**



Sol. Referring to the adjacent figure,

By constraint relation, $-v' = \frac{-v + v_p}{2}$

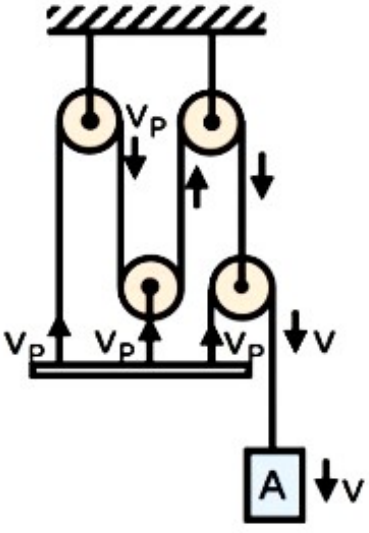
$\Rightarrow \quad v' = \frac{-v_p + v}{2}$

Also, $-v_p = \frac{v_p - v'}{2}$ $\rightarrow$ By constraint relation

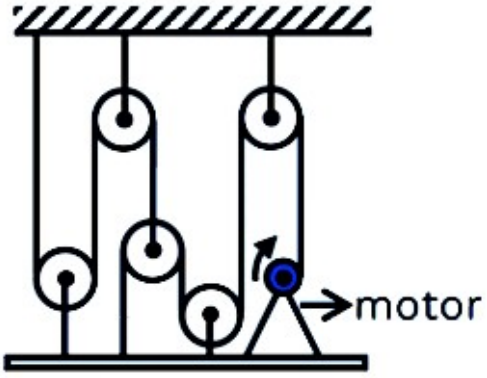
$-2v_p = v_p - v' \rightarrow v' = 3v_p$

So, $-v_p + v = 6v_p$

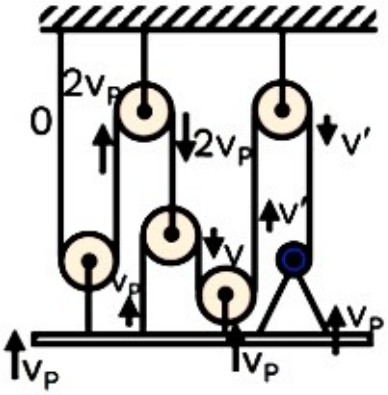
$7v_p = v \Rightarrow v_p = \frac{v}{7}$ upwards



Q. **Motor shown in the figure is wrapping the string at a rate of 10 m/s. Find the velocity of the plank.**



Sol. By constraint relation,



$$-2v_p = \frac{v_p - v}{2}$$

$$v = 5v_p$$

and $$-v_p = \frac{v - v'}{2}$$

$$-v_p = \frac{5v_p - v'}{2}$$

$$v' = 7v_p$$

Given that $v' + v_p = 10\text{m/s}$ ...

($\because$ $v' - (-v_p)$ = velocity of wrapping)

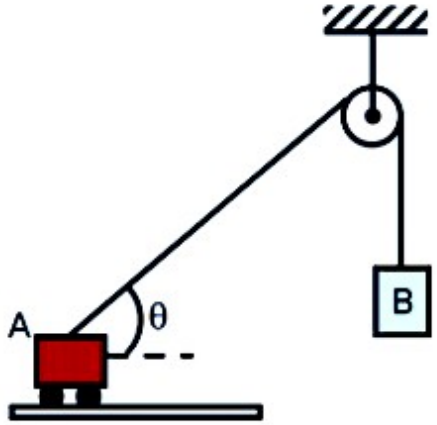
So, $7v_p + v_p = 10$

$$8v_p = 10$$

$$v_p = \frac{10}{8} = \frac{5}{4}\text{ m/s}$$

Constraint Relations for Variable Angles

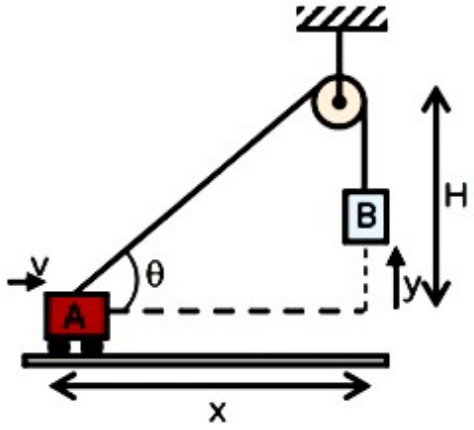
Q. **At the given instant, velocity of cart A is v towards right. Find velocity of block B.**



Sol. **Method-1**

⇒ String length constant method

$$L = (H - y) + \sqrt{(H^2 + x^2)}$$



Differentiate above expression w.r.t. t,

$$\frac{dL}{dt} = \frac{dH}{dt} - \frac{dy}{dt} + \frac{1}{2\sqrt{H^2 + x^2}}\left[\frac{2HdH}{dt} + \frac{2xdx}{dt}\right]$$

Now, L and H are constant, so $\frac{dL}{dt} = 0, \frac{dH}{dt} = 0$

$$\therefore \quad \frac{dy}{dt} = \frac{x}{\sqrt{H^2 + x^2}} \cdot \frac{dx}{dt}$$

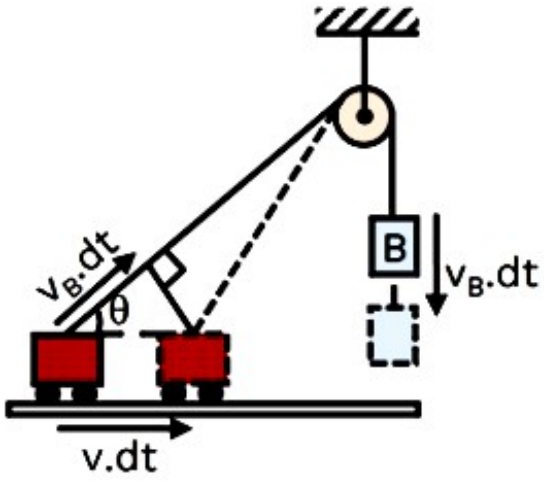
Now, $\frac{x}{\sqrt{H^2 + x^2}} = \cos\theta$

So, $v_B = v.\cos\theta$ $\quad \ldots\left(\because \frac{dy}{dt} = v_B, \frac{dx}{dt} = v_A\right)$

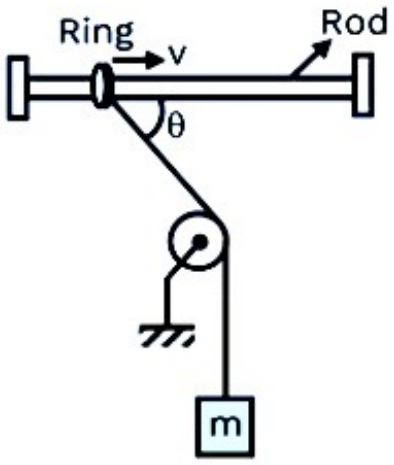
Method-2

$\cos\theta = \frac{v_B.dt}{v.dt}$

$v_B = v.\cos\theta$

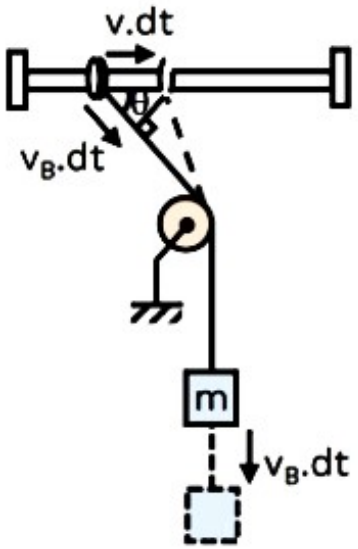


Q. **Ring is moving towards right with velocity v as shown. At the given instant, find velocity of block of mass m as shown.**

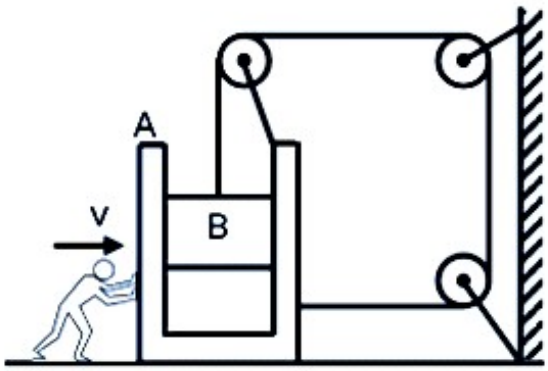


Sol. $\cos\theta = \frac{v_B.dt}{v.dt}$

$v_B = v\cos\theta$



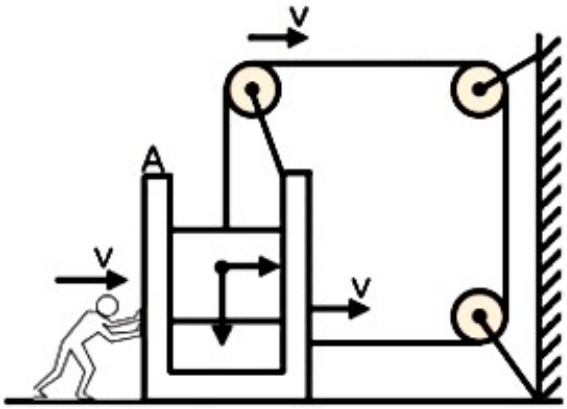
Q. **Find the speed of block B at given instant.**



Sol. $v_{B_y} = v + v = 2v$

$v_{Bx} = v$

Speed of block B, $v_B = \sqrt{v_{B_x}^2 + v_{B_y}^2}$



$= \sqrt{(2v)^2 + (v)^2}$

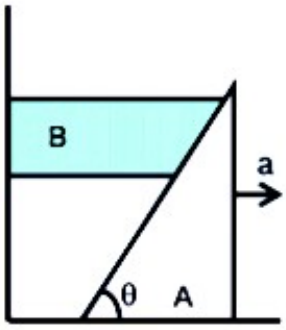
$= \sqrt{4v^2 + v^2} = \sqrt{5v^2}$

$= v\sqrt{5}$

Velocity of B, $\vec{v}_B = \left(v\hat{i} - 2v\hat{j}\right)$

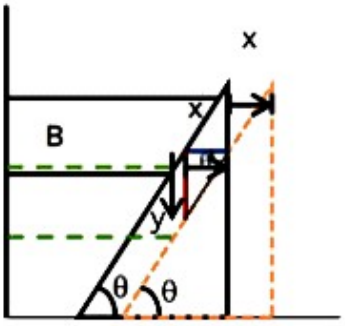
Wedge Constraint Relations

Q. **Find the acceleration of block B if acceleration of wedge A is equal to "a" towards right.**



Sol. To solve such kind of questions, we drop ink in the gap between block and wedge so that a stain of ink is formed on both.
Then, we simply displace one body and the other body displaces accordingly. We can measure the displacements by looking at the displacement of ink stain on both and find a suitable relation.

On displacing the wedge A towards right by x, the block B falls by distance y as shown.
Then, from figure,



$$\tan\theta = \frac{y}{x}$$

$$y = x.\tan\theta$$

Differentiating both sides wrt t,

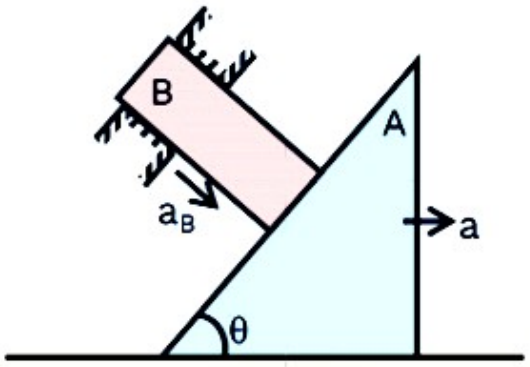
$$\frac{dy}{dt} = \frac{dx}{dt}.\tan\theta$$

Differentiating again,

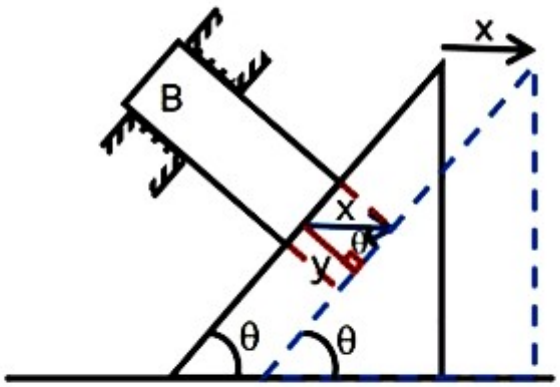
$$\frac{d^2y}{dt^2} = \frac{d^2x}{dt^2}.\tan\theta$$

$$a_B = a.\tan\theta$$

Q. **Find a_B, if wedge has an acceleration a.**



Sol. On displacing the wedge by x towards right, rod B moves by distance y.



From figure,

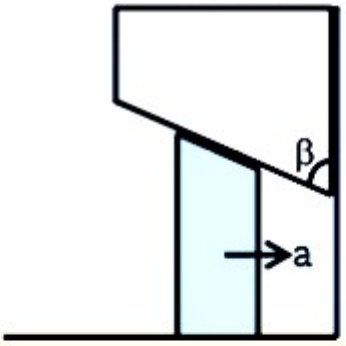
$$\sin\theta = \frac{y}{x}$$

$$y = x.\sin\theta$$

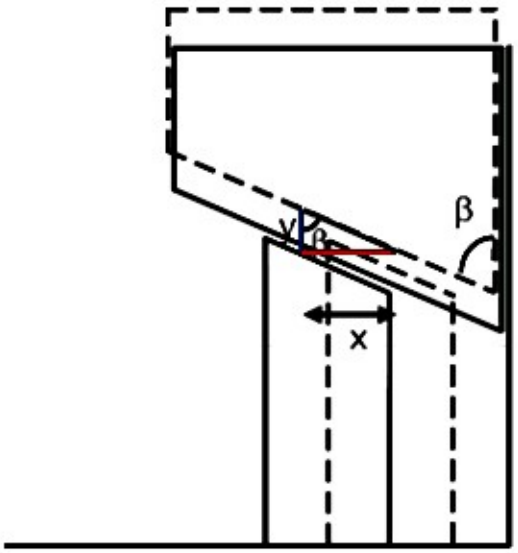
$$\Rightarrow \quad \frac{d^2y}{dt^2} = \frac{d^2x}{dt^2}.\sin\theta$$

$$a_B = a\sin\theta$$

Q. **Find the relation between accelerations of the given wedges.**



Sol. From figure,

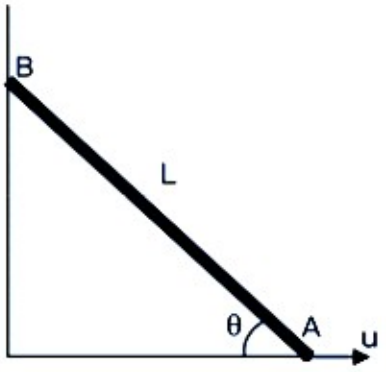


$$\tan\beta = \frac{x}{y}$$

$$y = \frac{x}{\tan\beta}$$

$$\Rightarrow \quad a_B = \frac{a}{\tan\beta}$$

Q. **A rod is sliding on smooth walls as shown. If velocity of end A is u, then find the velocity of end B.**

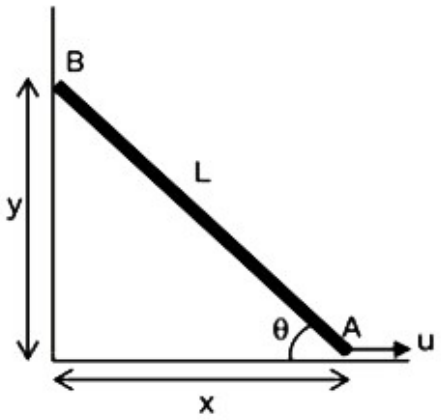


Sol. By Pythagoras theorem,

$$L^2 = x^2 + y^2$$

Differentiating both sides,

$$2L.\frac{dL}{dt} = 2x\frac{dx}{dt} + 2y.\frac{dy}{dt}$$



As L is constant, so $\frac{dL}{dt} = 0$

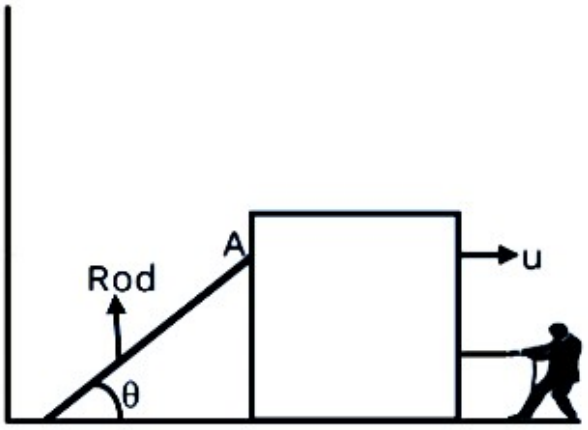
$$\therefore \quad \frac{dy}{dt} = -\left(\frac{x}{y}\right).\frac{dx}{dt}$$

Now, $\tan\theta = \frac{y}{x}$, $\frac{dy}{dt} = v_B$; $\frac{dx}{dt} = v_A$

So, $v_B = -\frac{v_A}{\tan\theta} = -\frac{u}{\tan\theta}$

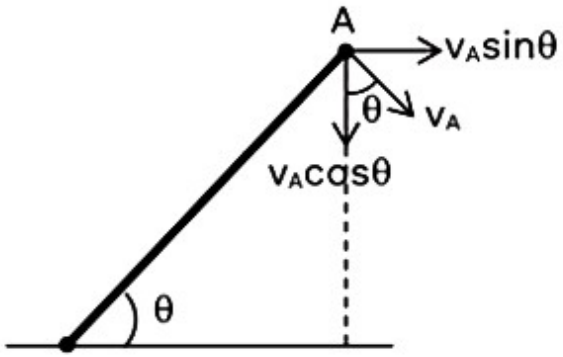
–ve sign indicates end B is moving downwards.

Q. **A rod is in contact with a block at given instant. The rod is hinged at other end. If a man pulls the block with speed u, find the velocity of end A of the rod.**



Sol. As the rod is hinged, so it will rotate about it. Then velocity of end A must be tangential.

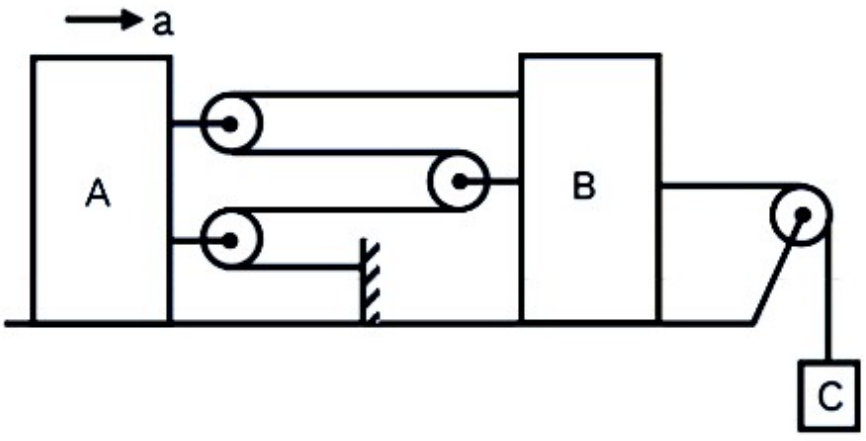
Also, the rod is always in contact with the block, so horizontal component of velocity of end A must be equal to the velocity of the block.



$$v_A \sin\theta = u$$

$$\Rightarrow \quad v_A = \frac{u}{\sin\theta} = u.\operatorname{cosec}\theta$$

Q. **If acceleration of block A is 'a' towards right, then find the acceleration of block C.**



Sol. For pulley P_4,

$a_B = a_C$

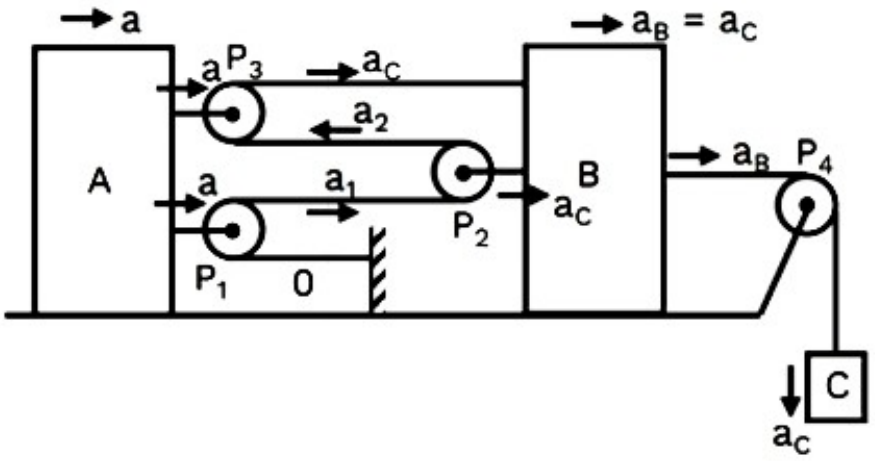
For pulley P_1,

$$a = \frac{0 + a_1}{2} \Rightarrow a_1 = 2a$$

For pulley P_2,

$$a_C = \frac{a_1 - a_2}{2}$$

$2a_C = 2a - a_2$, then, $a_2 = 2a - 2a_C$



and, for pulley P_3,

$$-a = \frac{a_2 - a_C}{2}$$

$\Rightarrow$ $-2a = 2a - 2a_C - a_C$

$3a_C = 4a$

$$a_C = \frac{4}{3}a$$

Free Body Diagram

A diagram drawn for a body by removing all supports and connections and replacing them by suitable forces is called free body diagram.

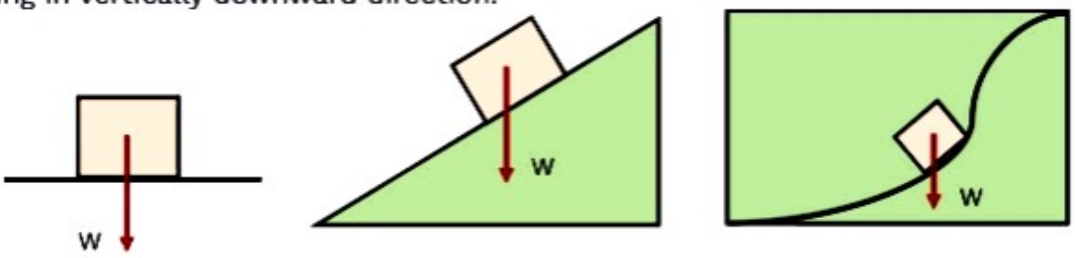
Steps to draw free body diagram

1. Isolate the object or the system from other objects by specifying a boundary.
2. Draw the weight of the body and other non-contact forces (if any) acting on the body.
3. Draw the contact forces like normal reaction, tension, spring force, applied force etc acting on the boundary of the body.

Common forces in mechanics

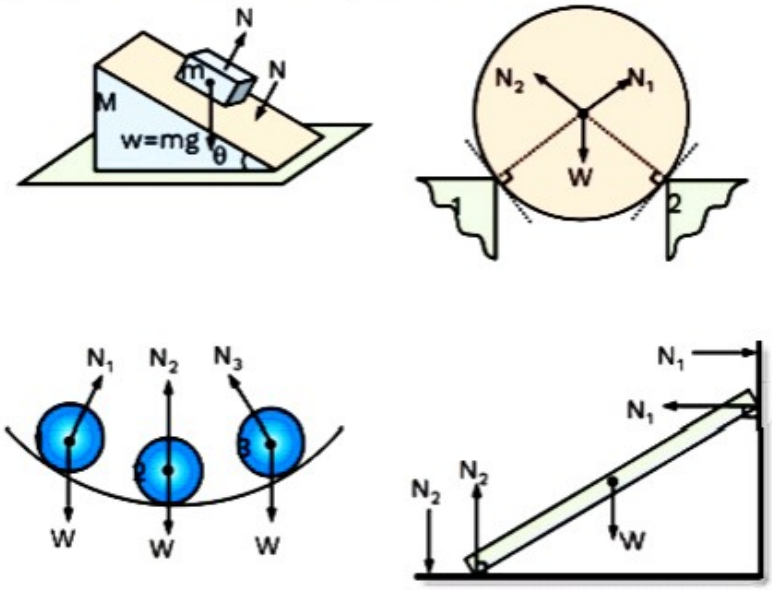
(a) Weight (W): The gravitational pull on a body due to earth towards its centre is termed as weight of the body. Magnitude of weight is given by W=mg and it is always directed vertically downwards (i.e., towards the centre of earth).

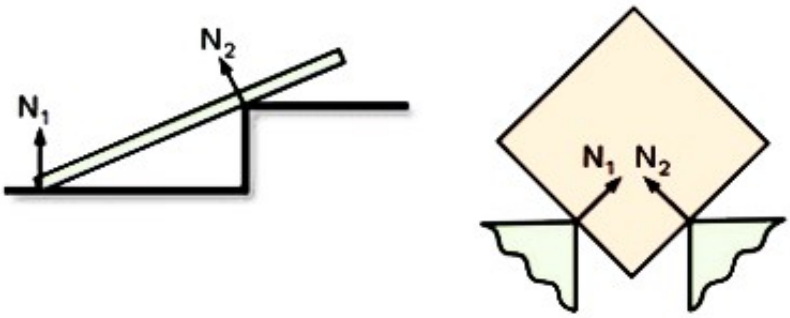
Some of the cases are shown in the figure below where the weight of a body is shown acting in vertically downward direction.



(b) Normal reaction (N): Normal reaction is a contact force which always acts in the direction perpendicular to the surfaces in contact. If there is no contact, then normal reaction will be zero.

Some of the cases are shown in the figure below where the normal reaction on a body is shown acting perpendicularly to the points of contact.





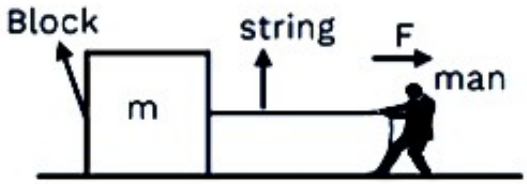
(c) Tension (T): Tension is the force generally occurring in strings and rods. It tries to elongate them. While drawing the free body diagram of a body, tension is always drawn acting away from the body.

Note: Properties of string

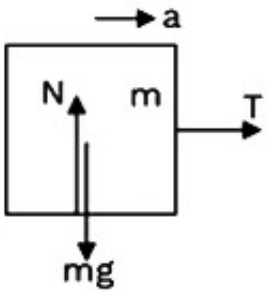
(a) A string pulls the bodies to which it is connected by a force equal to the tension in string.

(b) Tension is same at every point of a massless ideal single string.

Q. **Draw the FBD for the block, string and the man in the system given below.**



Sol. • **FBD of block**



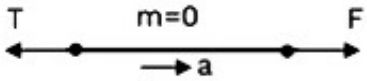
T is the tension in the string.

N = mg

T = ma ...(By 2nd Law)

(N and mg have same line of action. They are drawn separately just for clarity in figure.)

- **FBD of string**



$F - T = ma$

$F - T = 0(a)$

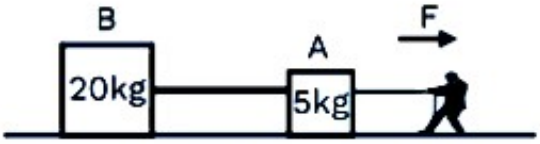
$F = T$

- **FBD of man**



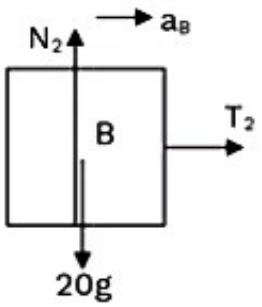
f is frictional resistance from ground.

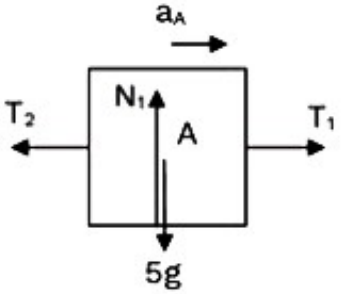
Q. **Draw the FBD of the blocks A and B shown in the figure.**



Also, find tension in strings and acceleration of blocks if F = 100 N.

Sol. FBD of A and B are shown below:





For string pulled by man,

$T_1 = F = 100N$

For block A,

$N_1 = 5g$

$T_1 - T_2 = 5a_A$

For block B,

$T_2 = 20\ a_B$

As the blocks are connected, so for their constraint motion,

$a_A = a_B = a$

So, $T_2 = 20a$

$T_1 - T_2 = 5a$

$100 - 20a = 5a$

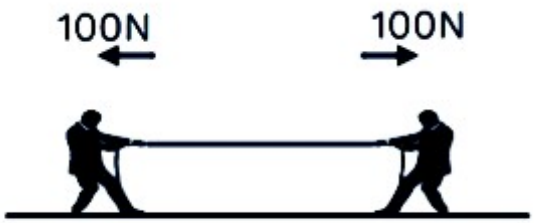
$25a = 100$

$a = 4\ ms^{-2}$

so, $T_2 = 20 \times 4 = 80N$

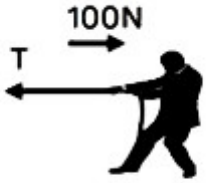
$T_1 = 100N$

Q. **Find the tension in the string shown below.**



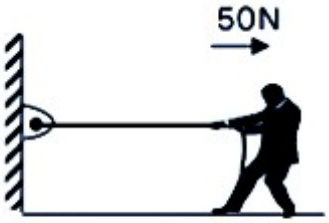
Sol. Draw the FBD of the string





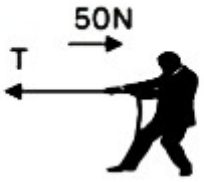
Tension, T = 100 N

Q. **Find tension in string.**



Sol. R is reaction force from hook on wall.





Tension, T = 50 N

(d) Spring force (F_s): A spring is a metallic coil which exerts a spring force when its natural length is changed. It pushes a body if its length is decreased from its natural length while it pulls a body if its length is increased from its natural length.

Equilibrium of coplanar forces

When a set of coplanar forces are acting on a body, then for the equilibrium of the body, net force on it must be zero.

The condition for equilibrium is thus given as,

$$\Sigma\vec{F} = 0$$

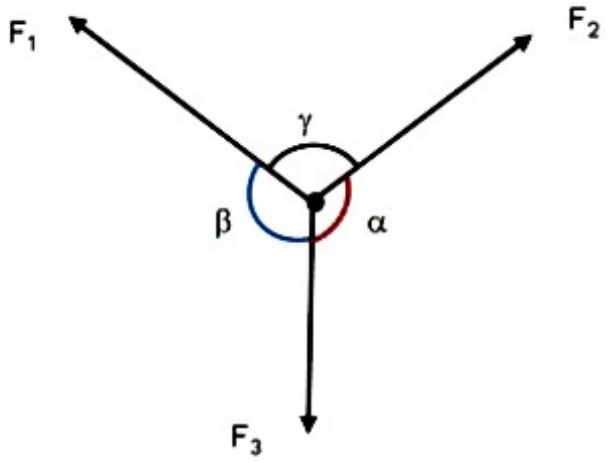
$$\vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \ldots\ldots\ldots = 0$$

We can also write for a set of planar forces in x-y plane,

$$\Sigma\vec{F}_x = 0 \text{ and } \Sigma\vec{F}_y = 0$$

Lami's theorem

Lami's theorem gives the condition for equilibrium for three concurrent forces. If three concurrent forces are acting on a body and the body is in equilibrium, then as per Lami's theorem, the relation between three forces and the angles formed is given as



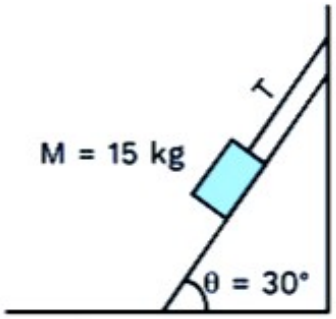
$$\frac{F_1}{\sin\alpha} = \frac{F_2}{\sin\beta} = \frac{F_3}{\sin\gamma}$$

Where, α = angle between $\vec{F}_2$ and $\vec{F}_3$

β = angle between $\vec{F}_3$ and $\vec{F}_1$

γ = angle between $\vec{F}_1$ and $\vec{F}_2$

Q. **A 15 kg block is tied to a string on an inclined plane (angle 30°). The tension T in the string is (Take g = 10 m/s²)**



Sol. Given that,

M = 15 kg,

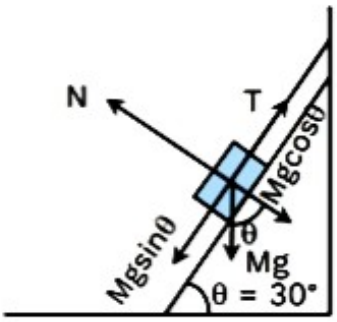
θ = 30°,

g = 10 m/s²

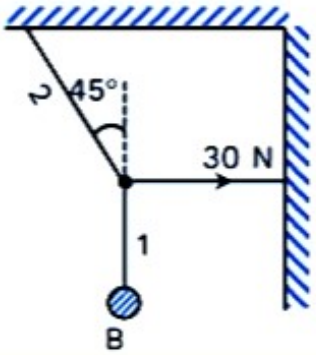
As the block is held by the string,

∴ $T = Mg\sin\theta = 15 \times 10 \times \sin 30°$

$$= 15 \times 10 \times \frac{1}{2} = 75\ N$$



Q. **In the given figure, find the weight of particle B is the tension in the horizontal string is 30 N.**



Sol. Let T_1 be the tension in string 1 and T_2 be the tension in string 2.

Resolving T_2 into two rectangular components.

In equilibrium,

$T_2\sin 45° = 30\ N$ ...(i)

$T_2\cos 45° = T_1$ ...(ii)

$T_1 = W$ ...(iii)

From (i), we get

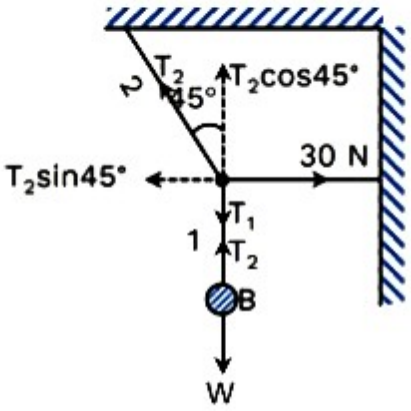
$T_2 = 30\sqrt{2}\,N$

On substituting this value of T_2 in (ii), we get

$$T_1 = 30\sqrt{2} \times \frac{1}{\sqrt{2}} N = 30N$$

From (iii), W = T_1 = 30 N

Hence, the weight of the body B is 30 N,



Q. **Three forces F_1, F_2 and F_3 together keep a body in equilibrium. If F_1 = 3 N along the positive x-axis, F_2 = 4N along the positive y-axis, then the third force F_3 is**

(A) 5 N making an angle $\theta = \tan^{-1}\left(\frac{3}{4}\right)$ with negative y-axis

(B) 5 N making an angle $\theta = \tan^{-1}\left(\frac{4}{3}\right)$ with negative y-axis

(C) 7 N making an angle $\theta = \tan^{-1}\left(\frac{3}{4}\right)$ with negative y-axis

(D) 7 N making an angle $\theta = \tan^{-1}\left(\frac{4}{3}\right)$ with the negative y-axis?

Sol. F_1 = 3 N along x-axis, F_2 = 4 N along y-axis

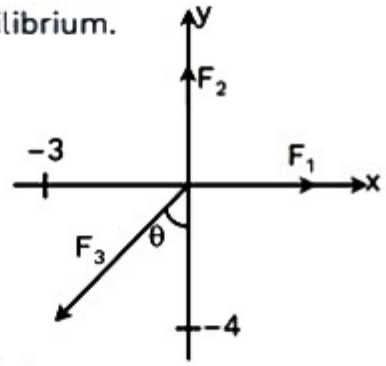
$\therefore \quad \vec{F}_1 = 3\hat{i}, \vec{F}_2 = 4\hat{j}$

Since three forces $\vec{F}_1$, $\vec{F}_2$ and $\vec{F}_3$ keep the body in equilibrium.

$\therefore \quad \vec{F}_1 + \vec{F}_2 + \vec{F}_3 = 0$

or $\quad \vec{F}_3 = -\left(\vec{F}_1 + \vec{F}_2\right) = -(3\hat{i} + 4\hat{j})$

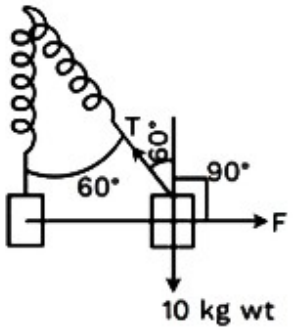
$$\left|\vec{F}_3\right| = \sqrt{3^2 + 4^2} = 5N$$



As it is clear from the figure, F_3 makes an angle $\tan^{-1}\left(\frac{3}{4}\right)$ with negative y-axis.

Q. **A mass of 10 kg is suspended from a spring balance. It is pulled aside by a horizontal string so that it makes an angle of 60° with the vertical. The new reading of the balance is**

Sol.



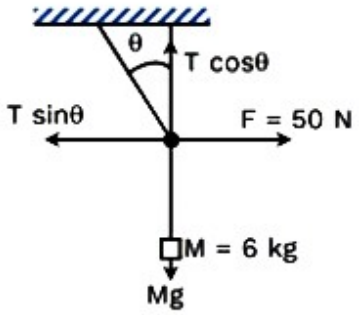
By Lami's theorem,

$$\frac{T}{\sin 90^\circ} = \frac{10\,\text{kg wt}}{\sin 150^\circ} \quad \text{or} \quad T = \frac{10}{\sin 30^\circ} = 20\,\text{kg wt}$$

Q. **A mass of 6 kg is suspended by a rope of length 2 m from a ceiling. A force of 50 N in the horizontal direction is applied at the midpoint of the rope. The angle made by the rope with the vertical, in equilibrium is**

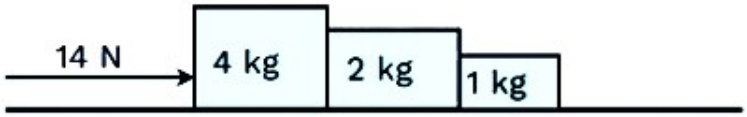
Sol. Let θ be the angle made by the rope with the vertical.
In equilibrium,
$T\sin\theta = 50\ N$
$T\cos\theta = (60\ kg)\ (10\ m/s^2) = 60\ N$

$$\therefore \quad \tan\theta = \frac{50N}{60N} = \frac{5}{6} \quad \text{or} \quad \theta = \tan^{-1}\left(\frac{5}{6}\right) = 40^\circ$$



Applications of Newton's Second Law and Constraint Relations

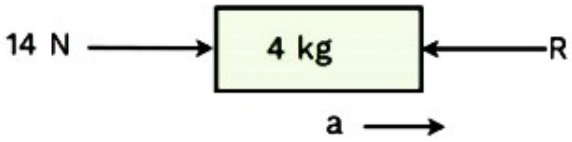
Q. **Three blocks of masses 4 kg, 2 kg, 1 kg respectively are in contact on a frictionless table as shown in the figure. If a force of 14 N is applied on the 4 kg block, the contact force between the 4 kg and the 2 kg block will be**



Sol. Common acceleration of the system,

$$a = \frac{14N}{4kg + 2kg + 1kg} = \frac{14}{7} ms^{-2} = 2ms^{-2} \qquad ...(i)$$

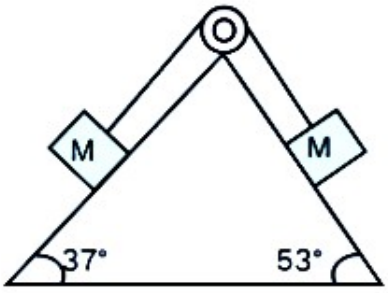
Let R be the contact force between 4 kg and 2 kg blocks. The free body diagram of 4 kg block is as shown in the figure.



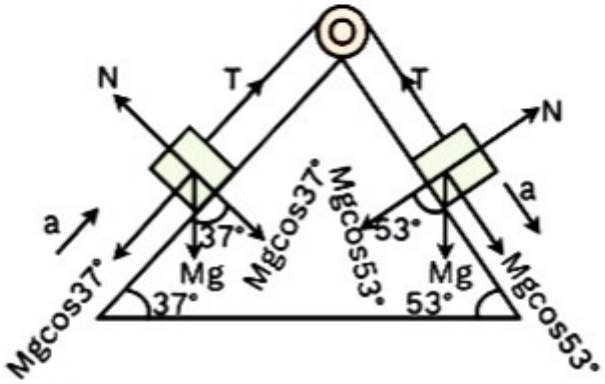
The equation of motion gives, $14 - R = 4a$

$$R = 14 - 4 \times 2 = 6 \text{ N}$$

Q. **The acceleration of system of two bodies over the wedge as shown in figure is**



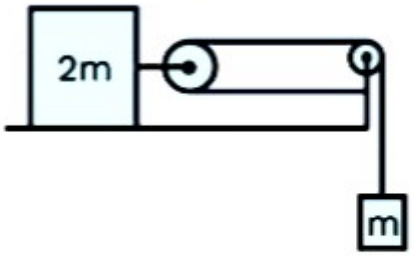
Sol.



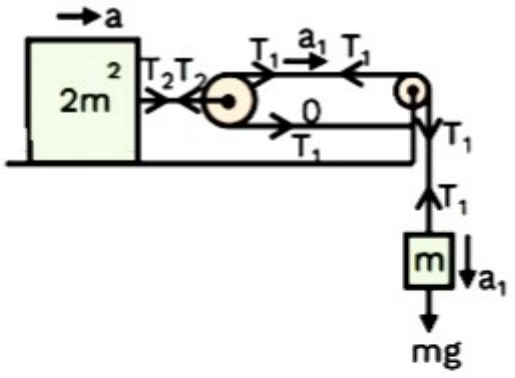
Let T be the tension in the string and a be acceleration of the system.
The equations of motion are
$Mg\sin53° - T = Ma$ ...(i)
and $T - Mg\sin37° = Ma$ ...(ii)
Adding (i) and (ii), we get

$$a = \frac{Mg\left(\sin 53^\circ - \sin 37^\circ\right)}{2M} = \frac{g}{2}\left(\frac{4}{5} - \frac{3}{5}\right) = \frac{g}{10} = 0.98 ms^{-2} \approx 1 ms^{-2}$$

Q. **Find the acceleration of the blocks shown in the system.**



Sol. Draw the system and represent the tensions in the two strings.



By constraint relation for the movable pulley,

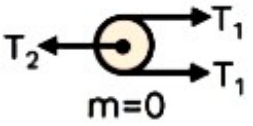
$$a_2 = \frac{0 + a_1}{2}$$

$$a_1 = 2a_2$$

As the movable pulley is massless,

So, $T_2 = T_1 + T_1$

$$T_2 = 2T_1$$



By Newton's 2nd Law,

for block of mass m, $mg - T_1 = ma_1$

for block of mass 2m, $T_2 = 2ma_2$

$$2T_1 = 2m\left(\frac{a_1}{2}\right)$$

$$T_1 = \frac{ma_1}{2}$$

So, $$mg - \frac{ma_1}{2} = ma_1$$

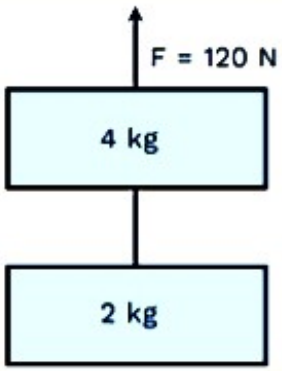
$$mg = \frac{3}{2}ma_1$$

$$a_1 = \frac{2g}{3}$$

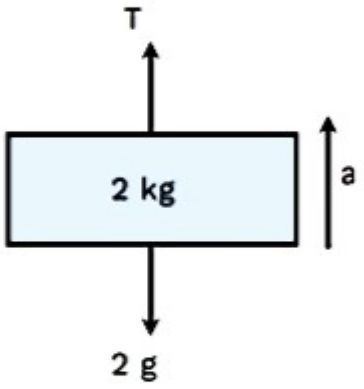
$$a_2 = \frac{a_1}{2} = \frac{g}{3}$$

Also, $T_1 = \frac{m}{2}\left[\frac{2g}{3}\right] = \frac{mg}{3}$ and, $T_2 = \frac{2mg}{3}$

Q. **Two blocks of masses 2 kg and 4 kg are attached by an inextensible light string as shown in the figure. If a force of 120 N pulls the blocks vertically upward, the tension in the string is**
(Take g = 10 ms^{-2})



Sol. Let a be common acceleration of the system and T be tension in the string.



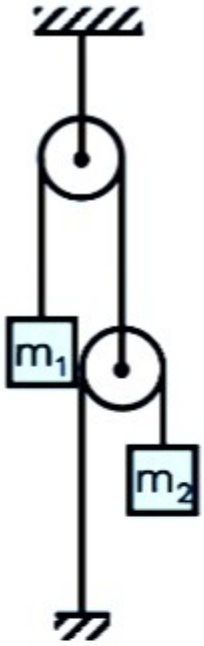
$$\therefore \quad a = \frac{F - 4g - 2g}{4 + 2} = \frac{120 - 40 - 20}{6} = 10 ms^{-2}$$

The free body diagram of 2 kg block is as shown in figure.

$\therefore$ $T - 2g = 2a$ or

$T = 2(a + g) = 2(10 + 10) = 40$ N

Q. **If $m_1 = 2kg$ and $m_2 = 3kg$, then find the acceleration of m_2.**



Sol. Represent accelerations and tensions in the strings.
By constraint relation for the lower movable pulley,

$$a_1 = \frac{0 + a_2}{2}$$

$$a_2 = 2a_1$$

By 2nd Law,

$$3g - \frac{T}{2} = 3a_2$$

and, $T - 2g = 2a_1$

$$T - 2g = 2\left[\frac{a_2}{2}\right]$$

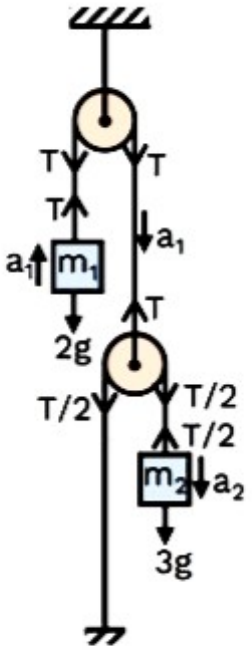
$$\Rightarrow \quad T = 2g + a_2$$

So, $$3g - \frac{1}{2}[2g + a_2] = 3a_2$$

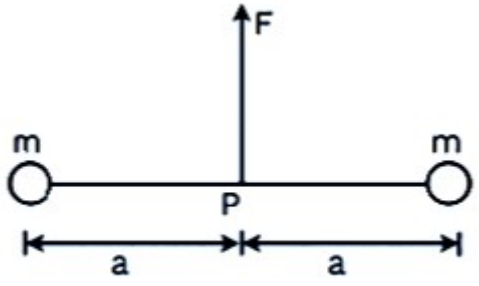
$$3g - g - \frac{a_2}{2} = 3a_2$$

$$2g = \frac{7}{2}a_2$$

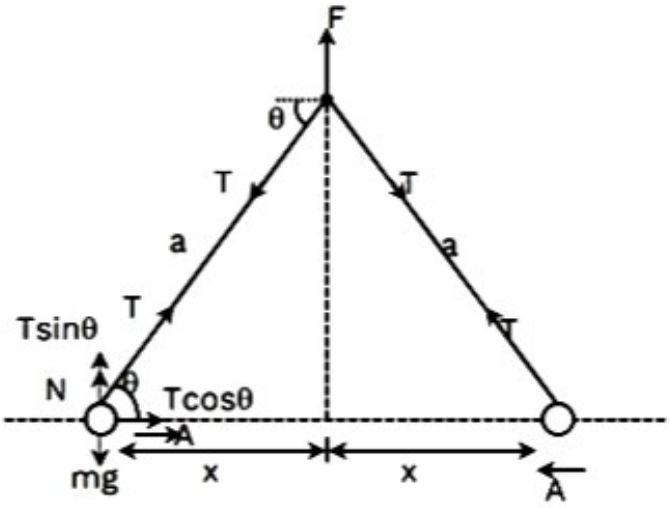
$$a_2 = \frac{4}{7}g$$



Q. **Two particles of mass m each are tied at the ends of a light string of length 2a. The whole system is kept on a frictionless horizontal surface with the string held tight so that each mass is at a distance a from the centre P (as shown in the figure). Now, the midpoint of the string is pulled vertically upwards with a small but constant force F. As a result, the particles move towards each other on the surface. The magnitude of acceleration, when the separation between them becomes 2x is**



Sol.



$F = 2T \sin\theta$

and, if A is the acceleration, then,

$T\cos\theta = mA$

$$A = \frac{F}{2m}\left(\frac{\left(\frac{x}{a}\right)}{\frac{\sqrt{a^2 - x^2}}{a}}\right)$$

$$\boxed{A = \frac{F}{2m}\frac{x}{\sqrt{a^2 - x^2}}}$$

- **Motion of a mass (monkey) along a string**

(a) **Case 1: When monkey is stationary on a string**. Let T be the tension in the string.



Draw the FBD of the monkey,



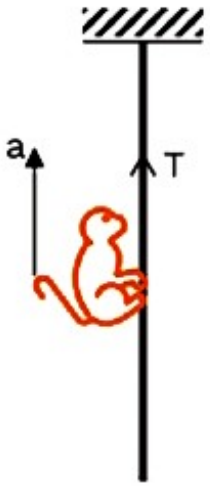
As the monkey is stationary, so, acceleration, a = 0

Then, by Newton's second law

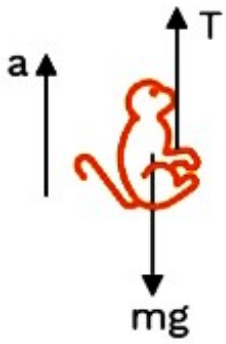
$T - mg = m(0)$

$T = mg$

(b) **Case 2: When monkey is accelerating upwards with acceleration a**



Draw the FBD of the monkey



$T - mg = ma$

Tension in the string, $T = m(g + a)$

Q. **If breaking load of string is 75N, mass of monkey is 4kg, find maximum upward acceleration of monkey.**

Sol. $T_{max} = m[g + a_{max}]$

$75 = 4[10 + a_{max}]$

$a_{max} = \frac{75}{4} - 10 = 18.75 - 10$

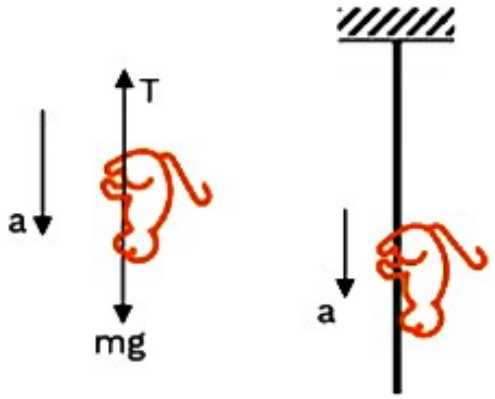
$a_{max} = 8.75 ms^{-2}$

(c) **Case 3: When monkey is accelerating downwards with acceleration a**

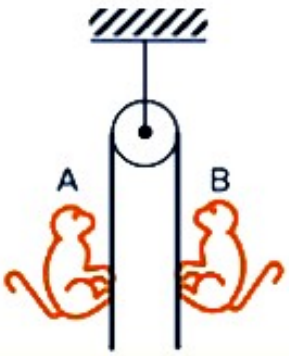
By Newton's Second law,

$mg - T = ma$

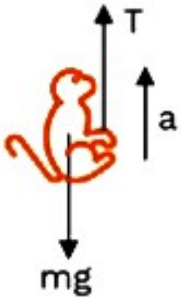
Tension, $T = m(g - a)$



Q. **Two monkeys of same mass are hanging as shown. If monkey A starts accelerating upwards with acceleration a, what will be the acceleration of monkey B.**



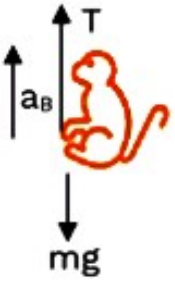
Sol. Draw FBD of monkey A,



$$T - mg = ma$$

$$T = mg + ma$$

Now, tension remains same in the whole string so, for monkey B,

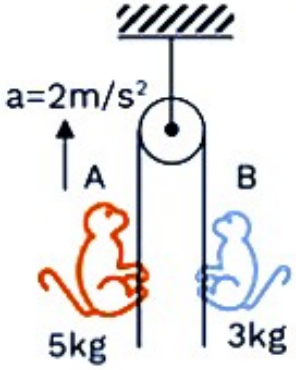


$$T - mg = ma_B$$

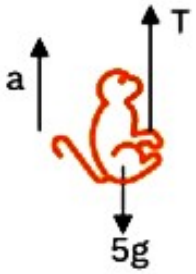
$$mg + ma - mg = ma_B$$

$$\text{So, } a_B = a$$

Q. **Find the acceleration of 3kg monkey in the system shown.**



Sol. For monkey A,

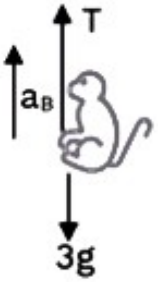


$$T - 5g = 5a$$

$$T - (5 \times 10) = (5 \times 2)$$

$$T = 60N$$

For monkey B,



$$T - 3g = 3a_B$$

$$60 - 30 = 3a_B$$

$$a_B = \frac{30}{3} = 10ms^{-2}$$

- **Concept of Weighing Machine :**

A weighing machine gives reading in kg.

The value of reading of a weighing machine is equal to the normal reaction on it divided by value of acceleration due to gravity (g).

Reading of weighing machine = $\frac{N}{g}$ (in kg)



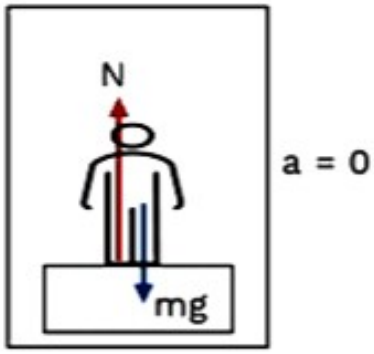
For the man,

$N = mg$

Reading of machine = $\frac{N}{g} = \frac{mg}{g} = m$ (in kg)

- **Weighing Machine in elevator (lift):**

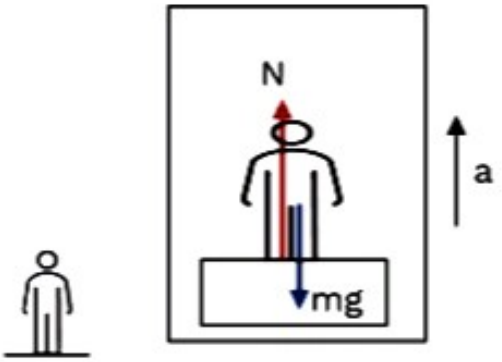
Case (a) : When the lift is stationary or moving with constant velocity:



For the man, $N = mg$ ...($\because a = 0$)

Reading of weighing machine = $\frac{N}{g} = \frac{mg}{g} = m$

Case (b) : When the lift is accelerating upwards with acceleration (a):



For an observer on ground, the man is accelerating upwards.

So, $N - mg = ma$

$N = m(g + a)$

Reading of weighing machine = $\frac{N}{g} = \frac{m(g+a)}{g}$

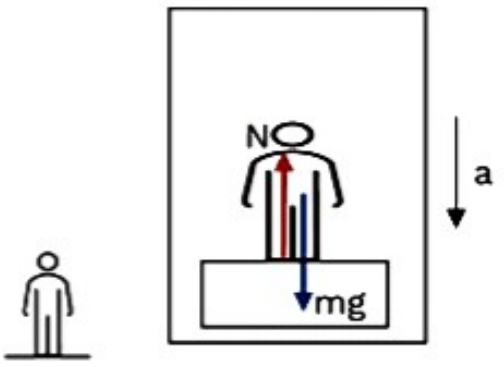
$= m\left[1+\frac{a}{g}\right]$

Case (c): When the lift is accelerating downwards with acceleration (a):

$mg - N = ma$

$N = m(g - a)$

Reading of weighing machine = $\frac{N}{g} = \frac{m(g-a)}{g} = m\left[1-\frac{a}{g}\right]$



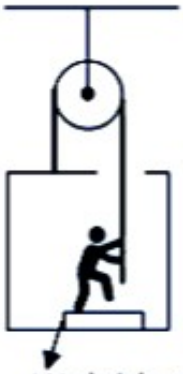
Then, for freely falling lift,

$a = g$

So, $N = m(g - g) = 0$

$\therefore$ Reading of weighing machine = $\frac{N}{g} = 0$. This condition is called "weightlessness".

Q. **Mass of box is 20kg and mass of man is 60kg in the given figure.**

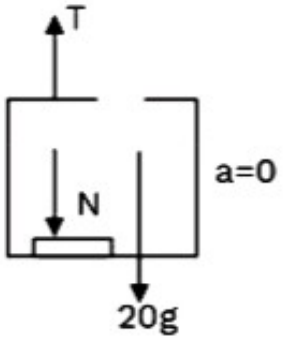


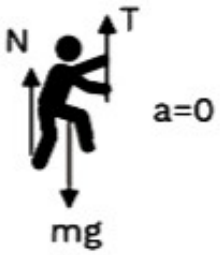
Weighing machine

(a) **If the man is stationary, find the tension in the string and the reading of the weighing machine.**

(b) **If the man wants to see his true mass (reading) in the weighing machine, then what should be the tension in the string.**

Sol. (a) Draw the FBD of man and lift,





For the lift, For the man,

$T = N + 20g$ $N + T = 60g$

So, $N + N + 20g = 60g$

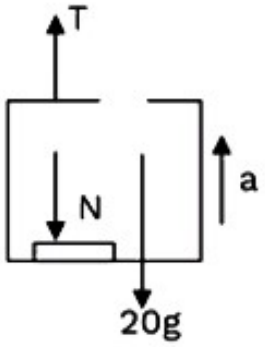
$2N = 40g$ then, $N = 20g$

Reading of weighing machine = $\frac{N}{g} = 20kg$

Tension, $T = 20g + 20g = 40g = 400N$

(b)





For the man,

$N + T - mg = ma$

For the lift,

$T - N - 20g = 20a$

For true reading, N should be 60g.

$\therefore \quad 60g + T - 60g = 60a$

and, $T - 60g - 20g = 20a$

So, $T = 60a$

Then, $60a - 80g = 20a$

$40a = 80g$

$a = 2g = 20\ m/s^2$

So, $T = 60 \times 20 = 1200\ N$

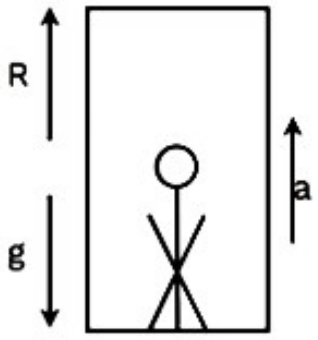
Q. **A lift moves vertically up with an acceleration a. Force exerted by a person of mass M on the floor of the lift is**

Sol. Using Newton's second law,

$Ma = R - Mg$

$R = M(g + a)$

$\therefore$ Force exerted by a person of mass M on the floor of the lift = $R = M(g + a)$



Q. **A person is standing on a weighing-scale and observes that the reading is 60 kg. He then suddenly jumps up and observes that reading goes to 70 kg. Then his maximum upward acceleration is?**

Sol. Before jumping the reading of the weighing scale on which the person is standing is 60 kg and after jumping on it, the reading goes to 70 kg.

If a be his maximum upward acceleration, then

$$(70\ \text{kg})g = (60\ \text{kg})g + (60\ \text{kg})a$$

$$\therefore \quad a = \frac{(70\text{kg})\,g - (60\text{kg})\,g}{60\text{kg}}$$

$$= \frac{g}{6} = \frac{9.8\text{m}/\text{s}^2}{6} = 1.63\ \text{m/s}^2$$

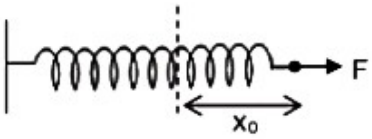
Cutting of Springs

SPRING:

Spring is a mechanical element which stores mechanical energy in form of spring potential energy. It is generally a helical coil of wire.



On fixing one end of the spring and pulling its other end gradually by applying a force, the coils get more separation between them. The total elongation in the spring is equally distributed between all the coils.



At equilibrium elongated position of the spring,

$F = Kx_0$, where K is spring constant.

For example, if 120 N of force is applied on a spring to elongate it by 10 cm, then,

$$F = 120\ N, \quad x_0 = 10\ cm = \frac{10}{100}m = \frac{1}{10}m$$

Then, spring constant, $K = \frac{F}{x_0} = \frac{120}{\left(\frac{1}{10}\right)} = 1200\,N/m.$

Smaller the length of a spring, higher is the value of K.

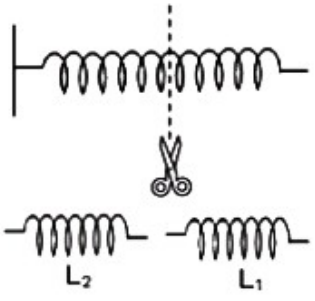
Larger the length of a spring, smaller is the value of K.

So, $K \propto \frac{1}{L_s}$, where L_s = Length of spring.

CUTTING OF SPRING:

We have a spring of length L having spring constant K.

Now if we cut the spring in the length ratio a : b, then,



$$\frac{L_1}{L_2} = \frac{a}{b} \Rightarrow L_1 = \frac{a}{b}L_2$$

and, $L_1 + L_2 = L$

$$\left(\frac{a}{b} + 1\right)L_2 = L$$

$$L_2 = \left(\frac{b}{a+b}\right)L$$

$$L_1 = \left(\frac{a}{a+b}\right)L$$

Now, $K \propto \frac{1}{L}$

So, $\frac{K_1}{K} = \frac{L}{L_1}$

$$K_1 = K\left[\frac{L}{L_1}\right] = K\left[\frac{L}{\left(\frac{a}{a+b}\right)L}\right] = \left(\frac{a+b}{a}\right)K$$

and, $\frac{K_2}{K} = \frac{L}{L_2}$

$$K_2 = K\left[\frac{L}{L_2}\right] = K\left[\frac{L}{\left(\frac{b}{a+b}\right)L}\right] \approx \left(\frac{a+b}{b}\right)K$$

Combination of Springs

(a) Series combination:

If two or more springs are connected such that force experienced by them is same, they are said to be connected in series.

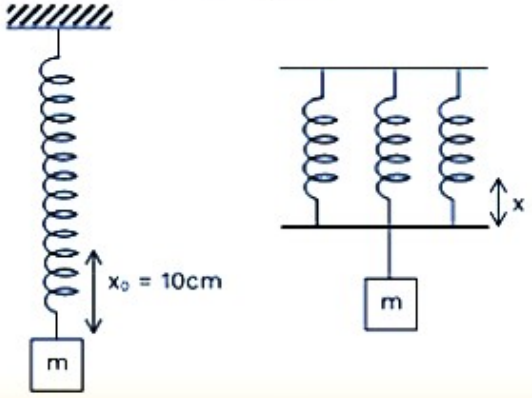
$$\frac{1}{K_{eq}} = \frac{1}{K_1} + \frac{1}{K_2} + \frac{1}{K_3} + \ldots$$

(b) Parallel combination:

If two or more springs are connected such that elongation or compression is same in both/ all, they are said to be connected in parallel.

Then, $K_{eq} = K_1 + K_2 + K_3 + \ldots$

Q. **A spring is hanging vertically and a mass m is attached to its lower end. As a result, spring elongates by 10 cm. Now, if the spring is cut in 3 equal parts and the same mass is attached to the combination as shown, what is the elongation in the combination of springs?**



Sol. If K is the spring constant of original spring,

Then, K (10 cm) = mg ...(i)

On cutting in 3 equal parts, spring constant of each cut part, K' = 3K

For parallel combination of 3 springs,

$$K_{eq} = K' + K' + K' = 3(3K) = 9K$$

Now, 9K (x) = mg ...(ii)

(ii) ÷ (i) gives,

$$\frac{9Kx}{K(10cm)} = \frac{mg}{mg}$$

$$9x = 10 \text{ cm}$$

$$x = \frac{10}{9} cm$$

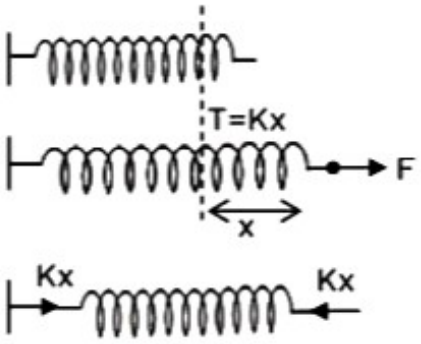
- **Property of Spring:**

Spring has tension in elongation just like the case of strings.

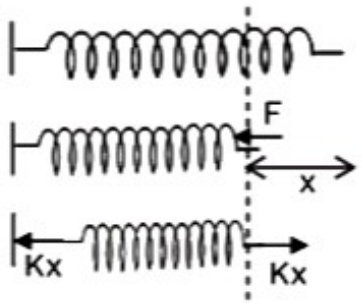
But unlike strings, spring can be compressed also.

If a spring is elongated by a length x from its natural length, then it has a tension

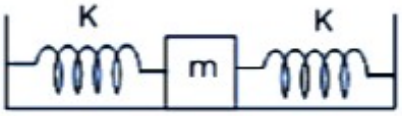
T = Kx and it pulls the bodies to which it is connected by a force equal to the tension or spring force (Kx).



If a spring is compressed by a length x from its natural length, then again it has a spring force Kx and it pushes the bodies to which it is connected by a force equal to the spring force (Kx).

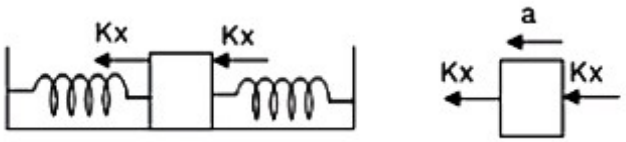


Q. **The two springs each of spring constant K = 200 N/m are initially in their natural lengths as shown.**



Now the block of mass m = 5kg is displaced towards right by 10 cm. Find acceleration of the block just after releasing it at this position.

Sol. On displacing the block slightly by x towards right, spring to the right compresses while spring to the left of the block elongates.
The elongated spring pulls the block towards it while the compressed spring is pushing it.

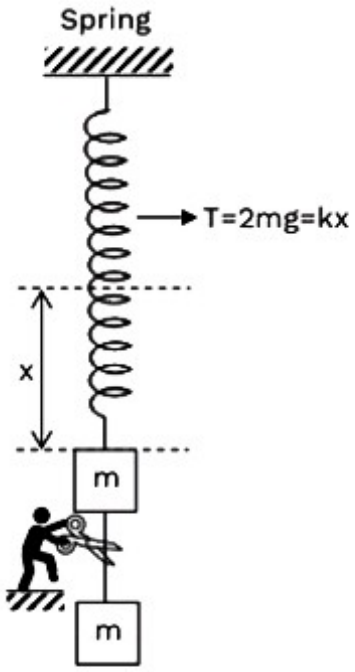


Net force on block, F = Kx + Kx = 2Kx ...(towards left)

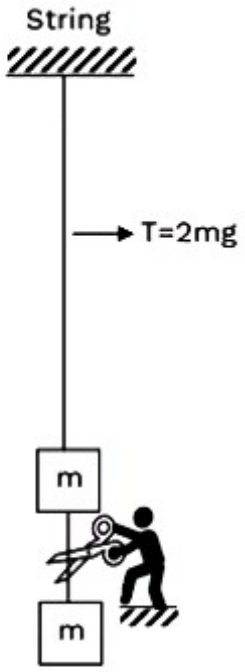
So, acceleration of the block at this instant,

$$a = \frac{F}{m} = \frac{2Kx}{m} = \frac{2 \times 200 \times \left(\frac{10}{100}\right)}{5} = 8 \text{ m/s}^2$$

Cutting of spring and string :



Just after cutting the lower string, tension in spring suddenly does not become zero.
It remains 2mg for few milli-seconds.

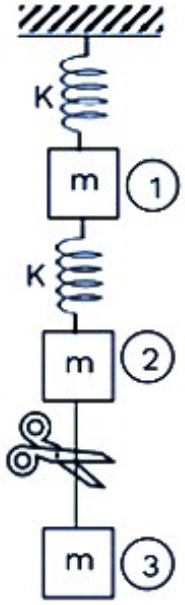


Initially, the tension in the upper string is 2mg. Just after cutting the lower string, tension in upper string suddenly becomes mg.

The time taken is due to the fact that if tension in spring has to become mg from 2mg, x has to be become $\frac{x}{2}$. So, it will take some time to reduce its elongation.

The sudden change in tension is because strings are inextensible, so it does not have to move to change its tension.

Q. **Find acceleration of block 1 just after cutting the lower string. Also find acceleration of 2 and 3.**



Sol. Just after cutting the lower string, spring force does not change suddenly.
So, acceleration of block 1, $a_1 = 0$

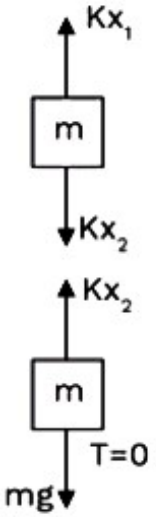
Acceleration of block 2, $a_2 = \left(\frac{Kx_2 - mg}{m}\right) \uparrow$

Acceleration of block 3,

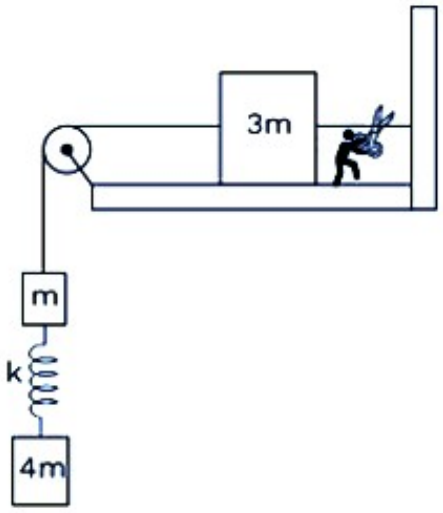
$a_3 = g$ (downwards $\downarrow$)

Initially, $Kx_2 = 2mg$

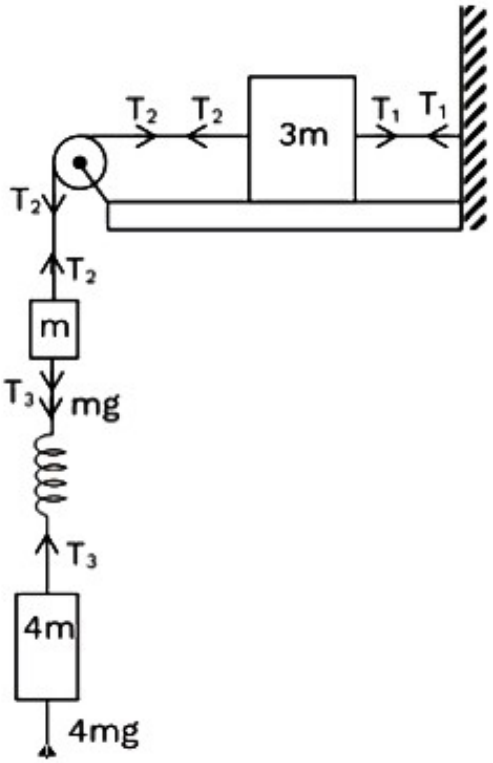
So, $a_2 = \frac{2mg}{m} - g = 2g - g = g$ (upwards $\uparrow$)



Q. **Find acceleration of each block just after cutting the string.**



Sol.



At initial equilibrium,

$T_3 = 4mg$

$T_2 = T_3 + mg = 4mg + mg$

$T_2 = 5mg$

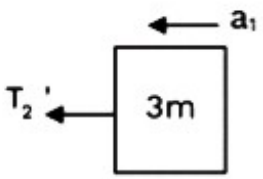
and, $T_2 = T_1$

So, $T_1 = 5mg$

Just after cutting the string, tension T_1 disappears suddenly, but tension in spring does not change suddenly. Tension T_2 in the string becomes T_2'.

For block of mass 3m,

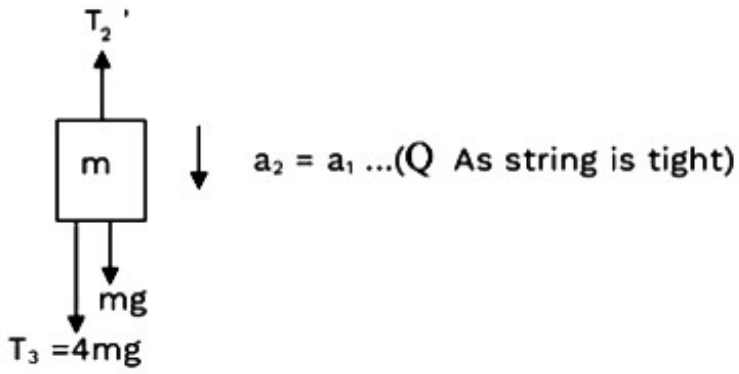
$$T_2' = 3ma_1$$



For block of mass m,

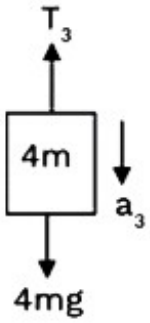
$$ma_2 = 4mg + mg - T_2'$$

$$ma_2 = 4mg + mg \quad T_2'$$



By second law, $ma_1 = 5mg - 3ma_1$

$$a_1 = \frac{5}{4}g = a_2$$

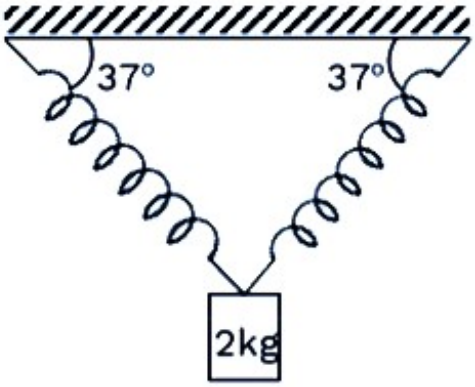


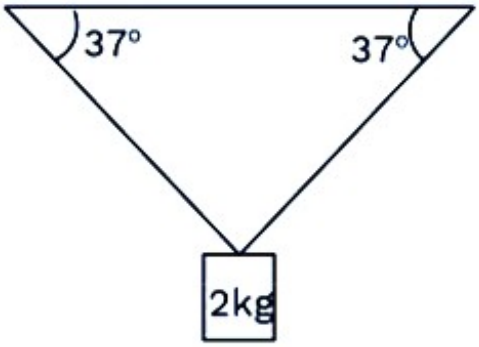
For block of mass 4m,

$4mg - 4mg = 4ma_3$

$a_3 = 0$

Q. **The blocks of mass 2 kg shown are in equilibrium. At t = 0, right spring in figure (i) and right string in figure (ii) breaks. Find the ratio of instantaneous acceleration of block in the two cases?**

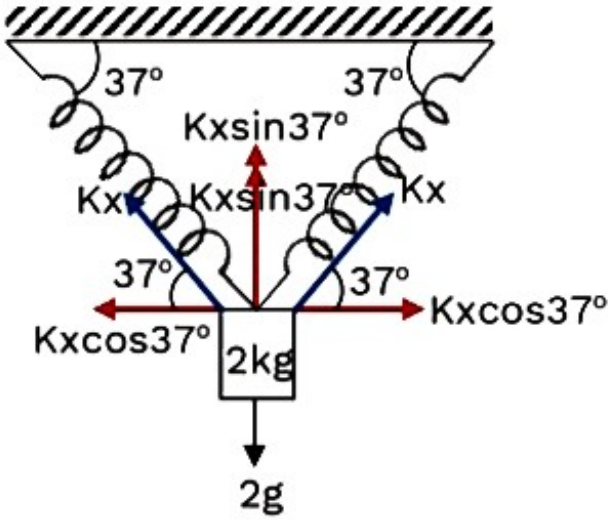




Sol. At initial equilibrium,

$$2K \times \sin 37^\circ = 2g$$

$$2K \times \left(\frac{3}{5}\right) = 2g$$



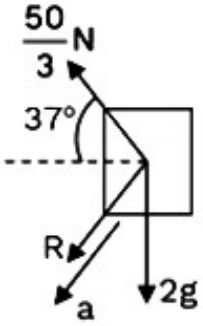
$$Kx = \frac{5g}{3} = \frac{50}{3}N$$

Just after cutting the right spring, Kx in right spring disappears but Kx in left spring remains equal to $\frac{50}{3}N$.

Resultant force acting on the block is,

$$R = \sqrt{20^2 + \left(\frac{50}{3}\right)^2 + 2(20)\left(\frac{50}{3}\right)\cos(90° + 37°)}$$

$$= \frac{50}{3}N$$



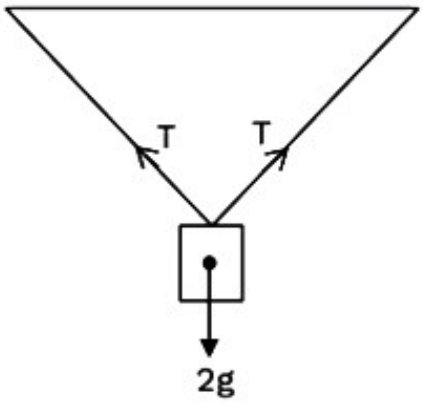
So, $a = \frac{R}{m} = \frac{50}{3\times 2} = \frac{25}{3} m/s^2$

For figure (ii),

At initial equilibrium,

$$2T\sin 37° = 2g$$

$$T = \frac{g}{\sin 37°} = \frac{50}{3}N$$



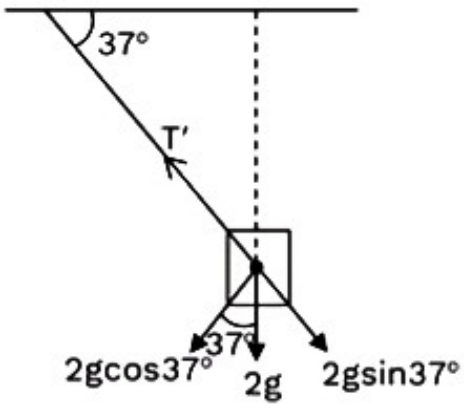
Just after cutting the right string, tension in left string changes to T'.

Now, block performs circular motion,

$$a_T = \frac{2g\cos 37°}{2} = g\cos 37°$$

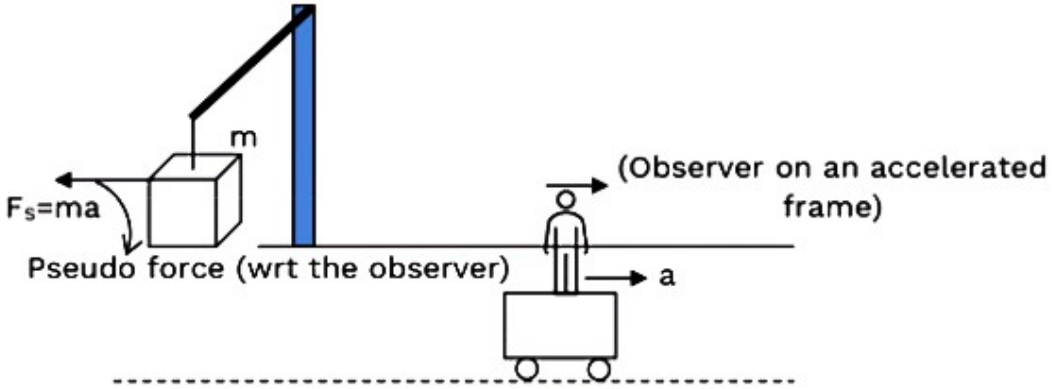
$$= 10\times\frac{4}{5} = 8 m/s^2$$

So, $\boxed{\frac{a}{a_T} = \frac{25}{3\times 8} = \frac{25}{24}}$

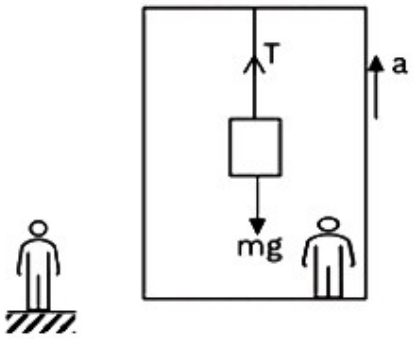


Pseudo Force

Pseudo force is a fictitious (imaginary) force which comes into play when motion of an object is being observed from an accelerated frame of reference (non-inertial frame).
The value of Pseudo force is equal to product of mass of body and acceleration of the reference frame.
The direction of Pseudo force is opposite to the direction of acceleration of reference frame.



An accelerated frame (non-inertial frame) can be converted into a non-accelerated frame (inertial frame) by applying Pseudo force.
Consider a block attached to a string hanging in an elevator.



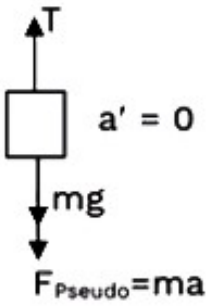
If lift is accelerating upwards, then for an observer on ground (inertial frame),

$T - mg = ma$

$T = mg + ma$

Now, for an observer inside the lift, the frame of reference is non-inertial. So, we have to add Pseudo force in the FBD of block as per the observer in the lift.

As per the observer in the lift, the acceleration of the block is zero.

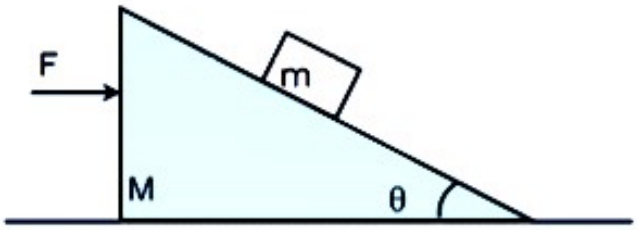


$T - mg - ma = ma'$

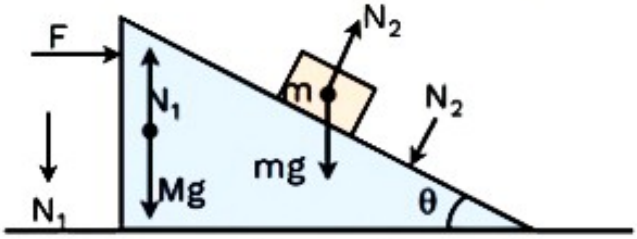
$T - mg - ma = m(0)$

$T = mg + ma$

Q. **In the figure below, a block of mass m is kept on a wedge of mass M and the combination is on a smooth surface. Find the applied force F on the wedge such that the block of mass m remains at rest on the wedge.**



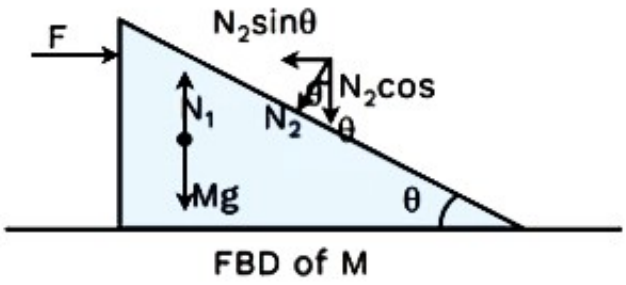
Sol. The forces acting on the system are shown in the figure below:



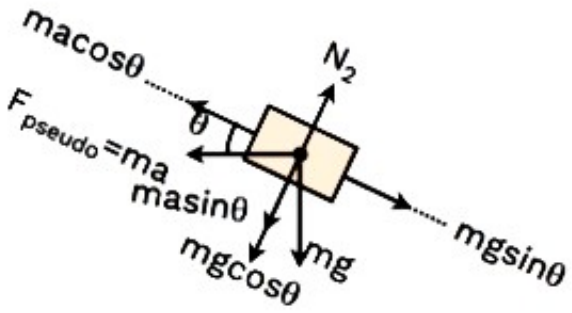
Forces acting on the system

As the block of mass m is at rest and the system moves on the frictionless surface as a single unit, so common acceleration, $a = \frac{F}{m+M}$

The free body diagrams of the wedge and the block can be drawn as,



FBD of M



A pseudo force (ma) is acting on the block from the frame of reference of the wedge. The direction of this pseudo force is opposite to the direction of acceleration of the wedge.

For the block of mass m to remain at rest,

$$mg \sin\theta = ma \cos\theta$$

$$\Rightarrow \quad a = g \tan\theta$$

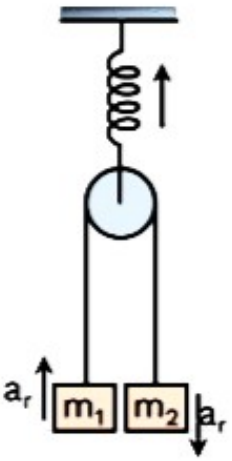
$$\Rightarrow \quad \frac{F}{m+M} = g\tan\theta$$

$$\Rightarrow \quad F = (m + M)g \tan\theta$$

Q. **An Atwood's machine having two blocks of masses m_1 = 1 kg and m_2 = 2 kg is suspended from a spring balance attached to ceiling of a lift moving upwards with an acceleration of $\frac{g}{10}$. Determine the reading of the spring balance.**

Take g = 10 ms^{-2}.

Sol. Let a_r be the relative acceleration of blocks with respect to pulley.
From the figure,



Absolute acceleration of $m_1 = a_r + a = a_r + \frac{g}{10}$ (upwards) and,

Absolute acceleration of $m_2 = a_r - a = a_r - \frac{g}{10}$ (downwards)
From Newton's 2nd Law,

$$T - m_1 g = m_1\left(a_r + \frac{g}{10}\right) \quad \ldots(1)$$

$$m_2 g - T = m_2\left(a_r - \frac{g}{10}\right) \quad \ldots(2)$$

Solving equation (1) and (2) with $m_1 = 1$ kg and $m_2 = 2$ kg, we get

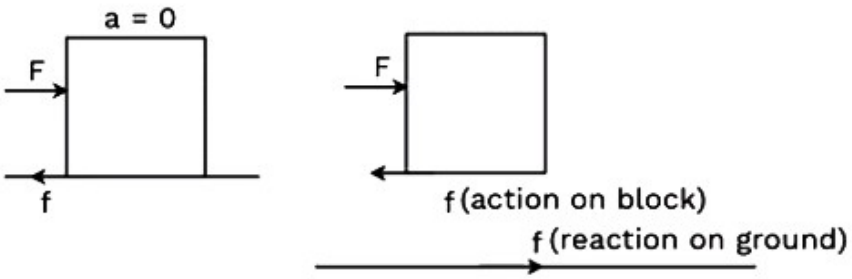
$$T = \frac{44}{3}N$$

The reading of the spring balance is therefore

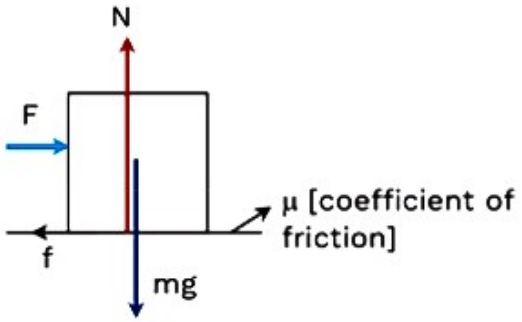
$$F = 2T = \frac{88}{3}N$$

Friction

Friction is a resistive force which opposes relative motion between two surfaces in contact.
It is a contact force, and it will only act if there is contact between surfaces having relative motion or trying to have relative motion.
It follows the action reaction law and friction force is electromagnetic in nature.



If a body is kept on a rough surface and it is at rest, then friction will be variable. This friction force is called static friction.
If a body is moving on a rough surface, then friction is constant and is called kinetic friction.



Consider a block of mass m kept on a rough surface of coefficient of friction μ.

Now start applying a horizontal force F varying its magnitude with time.

If the body is at rest, then at any instant,

Static friction force, f = Applied force, F

Value of static friction can vary from 0 to μN, where N = Normal reaction.

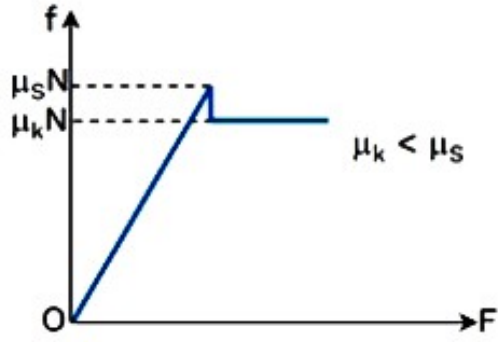
μN = Limiting friction

For applied force F > μN, friction becomes kinetic and constant.

$f_k = \mu_k N$,

where μ_k = coefficient of kinetic friction.

If the applied force F varies gradually, then friction (f) and applied force (F) variation can be plotted as



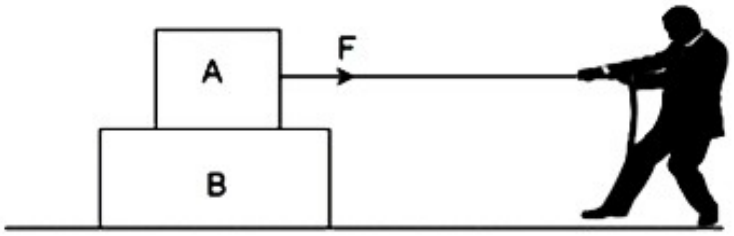
$$\text{Force of friction} \Rightarrow \begin{cases} 0 \le f \le \mu_s N \\ f_k = \mu_k N \end{cases}$$

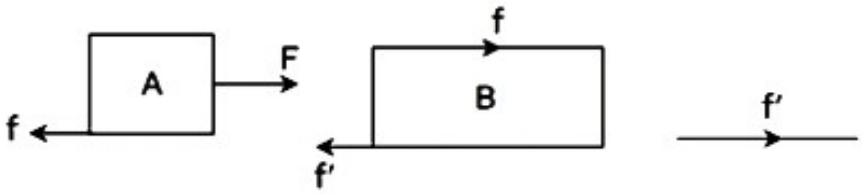
Kinetic friction is slightly less than limiting friction.

Static friction is self-adjusting in nature.

Maximum value of static friction is $\mu_s N$.

Friction always tries to stop "relative motion" not "motion". It can create motion also in order to stop relative motion.



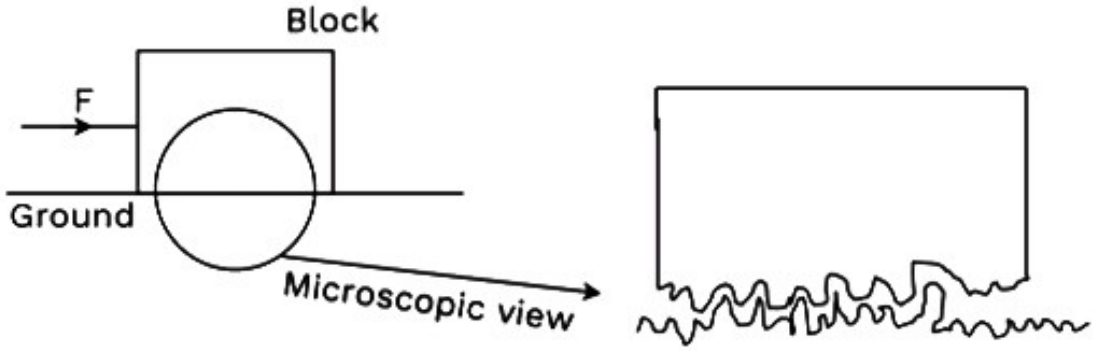


Here, in above condition, friction between A and B is creating motion for B in order to avoid relative motion between A and B.

- Value of μ is always positive and it can have any value between 0 to ∞.
- Value of μ depends upon nature of surfaces in contact.

Origin of Friction

Consider a block kept on a rough surface as shown.



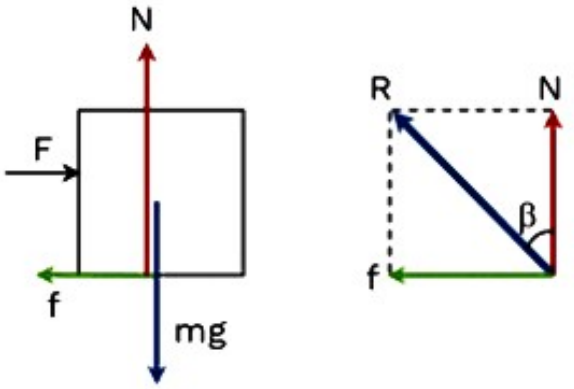
The surfaces appearing flat to naked eye have surface irregularities at microscopic levels. When the block is made to move on the surface by applying force, these surface irregularities interlock with each other causing an opposing force to the applied force. This interlocking causes frictional force.

Once the motion starts, the interlocking breaks. Due to this reason, the value of kinetic friction is lesser than static friction slightly.

ANGLE OF FRICTION:

Angle of friction is the angle between net contact force and the normal reaction from the surface.

Net contact force is the resultant of normal contact force and frictional contact force.



Net contact force, $R = \sqrt{N^2 + f^2}$

Angle between $\vec{R}$ and $\vec{N}$ is the angle of friction.

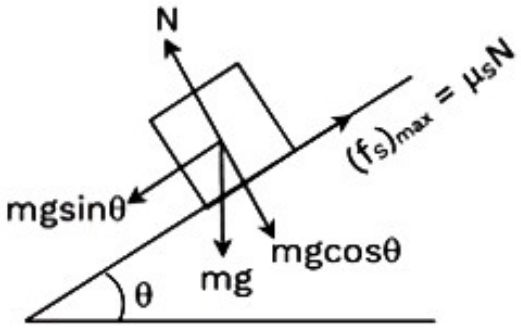
Angle of friction (β) must increase with increase in value of friction force.

CALCULATING VALUE OF μ FOR TWO BODIES:

If we have a block and a flat surface, we can easily find value of μ for this combination.

Take the flat surface, hinge its one end and start lifting it from the other end. Keep the block on the surface.

There will be a value of angle θ at which the block is just about to slide.



Measure that angle θ.

At limiting condition,

$mg \sin\theta = \mu_s N$

$mg \sin\theta = \mu_s\, mg \cos\theta$

$$\mu_s = \frac{\sin\theta}{\cos\theta} = \tan\theta$$

So, tan of the measured angle (θ) gives the value of coefficient of static friction. This angle (θ) is called **"Angle of Repose"**.

Q. **A block is kept on an inclined plane of inclination $\theta = 30°$. If $\mu = 0.75$, find the acceleration of block?**

Sol. If ϕ is angle of repose,

then, $\mu = \tan\phi$

$0.75 = \tan\phi$

$$\frac{3}{4} = \tan\phi$$

$$\phi = \tan^{-1}\left(\frac{3}{4}\right) = 37°$$

So, sliding will not start until θ is less than 37°.

As $\theta = 30°$ is less than 37°, so there will be no slipping.

$\therefore$ Acceleration of block, $a = 0$.

- **CALCULATION OF COEFFICIENT OF KINETIC FRICTION:**

If for inclination $\theta = \theta_0$, slipping starts, then $\tan\theta_0 = \mu_s$

Now, for $\theta > \theta_0$, slipping starts and block moves down the plane.

Measure the distance L and measure time t (using a stopwatch) which the block takes to reach the bottom.

Then, $a = g\sin\theta - \mu_k g\cos\theta$

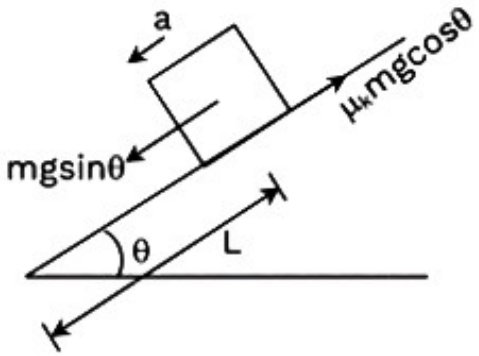
By equation of motion, $L = \frac{1}{2}at^2$

So, $\frac{2L}{t^2} = g[\sin\theta - \mu_k\cos\theta]$

We know the values of L, t, g and θ.

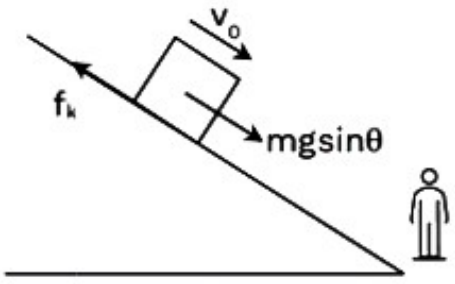
On putting these values, we can find value of μ_k.

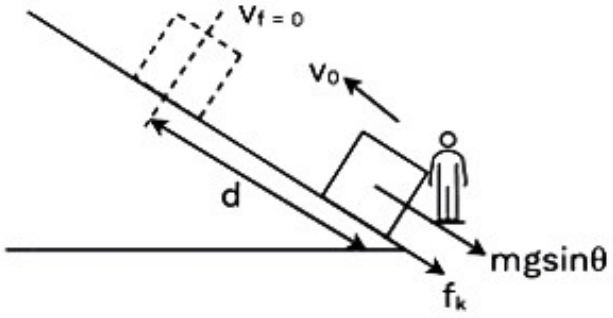
μ_k is always less than μ_s.



Q. **A man observes a block sliding on an inclined plane with a constant velocity v_0. As soon as the block reaches the bottom, the man picks the block and projects it upwards the inclined plane with same velocity v_0.**
Find the distance covered by the block before it comes to rest from the point of projection.

Sol.





As the block is sliding down the plane with constant velocity, so

$a = 0$

$mg\sin\theta - f_k = m(0)$

$f_k = mg\sin\theta$

When projected up the plane, f_k and $mg\sin\theta$, both act down the plane.

So, net retardation during upward journey,

$$a = -\frac{[f_k + mg\sin\theta]}{m}$$

$$= -\frac{[mg\sin\theta + mg\sin\theta]}{m}$$

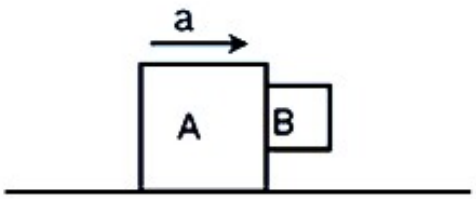
$$a = -2g\sin\theta$$

Now, by equation of motion, $v^2 - u^2 = 2as$

$$0^2 - v_0^2 = 2(-2g\sin\theta)d$$

So, $$d = \frac{v_0^2}{4g\sin\theta}$$

Q. **The value of friction coefficient between the two blocks A and B shown in the figure is 0.5. Find the minimum acceleration of block A so that the block B having a mass 5 kg doesn't fall from the surface.**



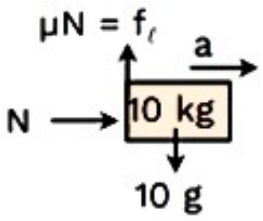
Sol. Pseudo force on the block B causes a normal reaction on it.

So, $N = ma$

For the block to not slip down,

$\Rightarrow$ $mg \le \mu N$

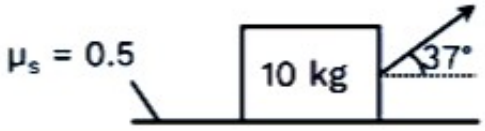
$\Rightarrow$ $mg \le \mu\, ma$



$\Rightarrow$ $a \ge \frac{g}{\mu}$

$\Rightarrow$ $a_{min} = \frac{g}{\mu} = \frac{10}{0.5} = 20\,ms^{-2}$

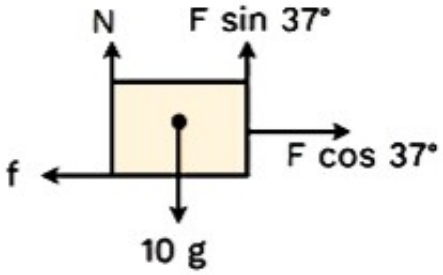
Q. **In the given figure, the value of the externally applied force F being increased gradually from zero. State whether the block will first slide from the surface or will be lifted upwards.**



Sol. **Case 1:** For the block to start lifting up in vertical direction,

$$F \sin(37°) + N - 10\,g \geq 0$$

N becomes zero just before lifting, so



$$\text{Lifting force, } F_{lift} \geq \frac{10g}{\left(\frac{3}{5}\right)}$$

$$\Rightarrow \quad F_{lift} \geq \frac{500}{3}N$$

Case 2: For block to start sliding in horizontal direction,

$$F \cos(37°) \geq \mu_s N$$

Since $N = 10g - F \sin(37°)$

$$\Rightarrow \quad F \cos(37°) \geq 0.5\,(10g - F \sin(37°))$$

$$\text{Hence } F_{slide} \geq \frac{50}{\cos(37°) + 0.5\sin(37°)}$$

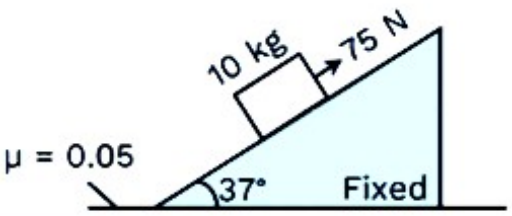
$$\Rightarrow \quad F_{slide} \geq \frac{500}{11}N$$

$$\text{Since } F_{lift} \geq \frac{500}{3}N$$

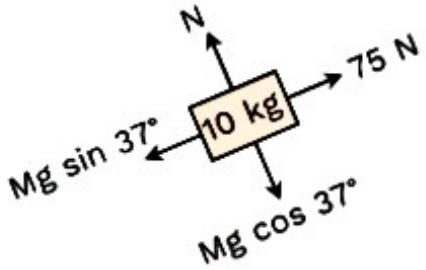
$$\Rightarrow \quad F_{slice} < F_{lift}$$

So, the block slides first instead of being lifted up.

Q. **In the given arrangement, find the acceleration of the block assuming that the block is initially kept at rest.**
Find the value of frictional force acting between the block and inclined plane.
Also find the additional force (excess of 75 N) for which the block starts to move up the inclined plane.
Finally, find the minimum force that can replace 75 N so that the block does not move.



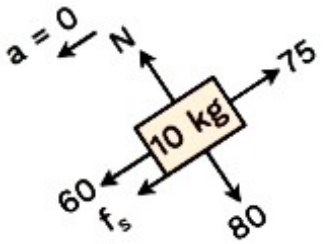
Sol. The free body diagram of the block excluding the frictional force can be drawn as,



Normal reaction, N = 10 g cos (37°) = 80 N

Now, $Mg\sin(37°) = (100)\left(\frac{3}{5}\right) = 60\,N < 75N$

So, net force on the block has a tendency to move the block up the inclined plane and thus the friction acts down the inclined plane.
Also, limiting friction, $f_{max} = \mu N = 40$ N



But $(F_{applied})_{net} = 75 - 60 = 15\ N < f_{max}$
So, block will not move, i.e., a = 0 and hence static friction acts between the block and the inclined plane.

Now the block is at rest or in equilibrium with respect to the inclined plane, so we have

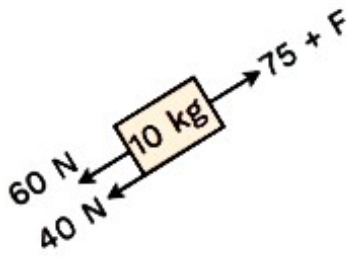
$$f_s + 60 = 75$$

$$\Rightarrow \quad f_s = 15 \text{ N}$$

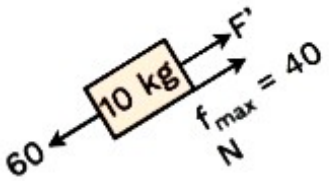
Let F be the additional force for which the block starts moving up the inclined plane, then we have

$$75 + F = 60 + 40$$

$$\Rightarrow \quad F = 25 \text{ N}$$



When 75 N is being replaced with another force F' , the block has a tendency to move downwards, so the friction acts up the inclined plane.



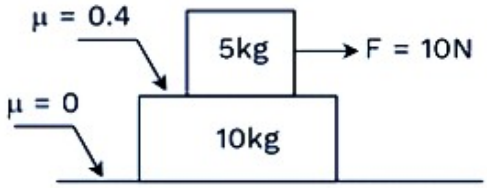
$$\Rightarrow \quad F' + 40 = 60$$

$$\Rightarrow \quad F' = 20 \text{ N}$$

Friction: Two Block System

We will discuss important problems related to friction in two block system in this section.

Q. **Consider the two-block system as shown below.**



Will the two blocks move together? If yes, find accelerations of blocks.

Sol. Let us assume first that both the blocks are moving together with common acceleration 'a'.

Then, $a = \frac{F}{5+10} = \frac{10}{15} = \frac{2}{3}\ m/s^2$

Draw the FBD of the blocks separately, then



Now, for 10 kg block, $f = 10 \times a = \frac{10 \times 2}{3} = \frac{20}{3} N$

Also, maximum possible value of f is, $f_{max} = \mu \times 5 \times g = 0.4 \times 5 \times 10 = 20\ N$

As $f < f_{max}$ $\left(\text{i.e.,} \frac{20}{3} < 20\right)$,

So, it is correct to assume that both blocks are moving together.

Hence, acceleration of the blocks is $\frac{2}{3} m/s^2$ and friction force between the blocks is $f = \frac{20}{3} N$.

Q. **In the previous problem, if applied force F = 90N, find the acceleration of the blocks and friction force between them.**



Sol. Assume both blocks are moving together,

then, $a = \frac{90}{5+10} = 6\,m/s^2$



For the 10kg block,

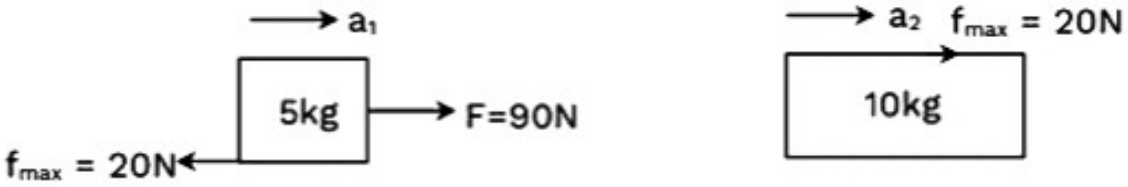
$f = 10 \times a = 10 \times 6 = 60N$

But $f_{max} = \mu \times 5g = 0.4 \times 5 \times 10 = 20N$

So, f cannot be greater than 20N. But here f is coming out to be 60N. So, our assumption is false and both blocks cannot move together.

Then, both are moving separately, and maximum friction is acting between the blocks.

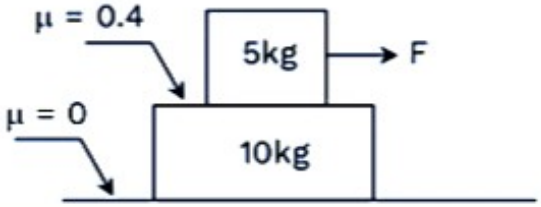
So,



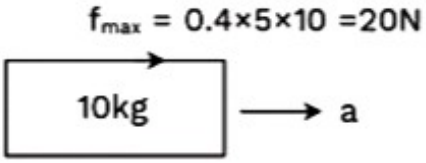
Then, $a_1 = \frac{90-20}{5} = \frac{70}{5} = 14\,ms^{-2}$.

$a_2 = \frac{20}{10} = 2\,ms^{-2}$

Q. **Find maximum value of force F so that both the blocks can move together.**



Sol. Both the blocks will be moving together until the upper and lower blocks are having friction force equal to limiting value.

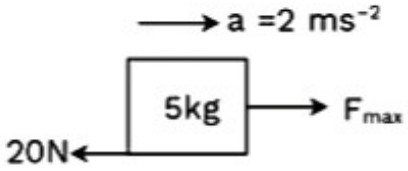


For the plank of mass 10 kg,

$20 = 10 \times a$

$a = 2\ ms^{-2}$

For the 5kg block,



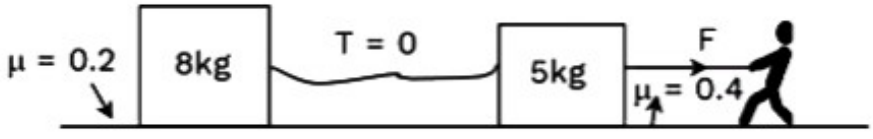
$F_{max} - 20 = 5 \times 2$

$F_{max} = 30N$

So, both the blocks can move together up to a maximum value of 30 N for force F.

FRICTION BREAKING :

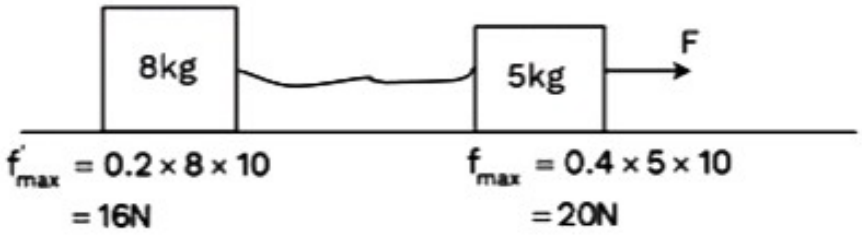
We have a system as shown. The string between the blocks is slacked initially and it has zero tension.



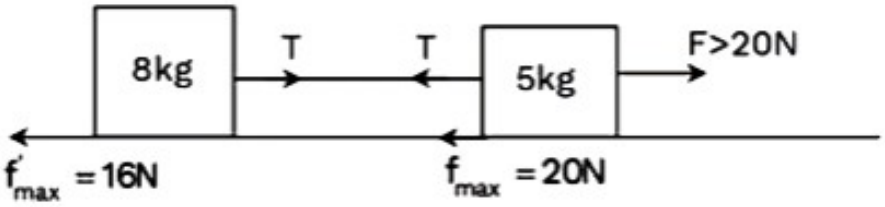
For what value of F, string will be taut?

For what value of F, both blocks are just about to move?

If F = 30N, what will be the value of frictional force on 8kg block?



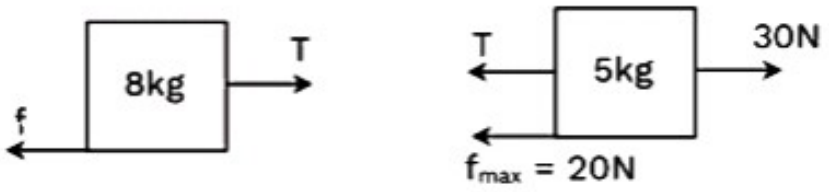
For string between blocks to become tight, F should be greater than 20N. For F just greater than 20N, string becomes tight.



For 8kg block to start motion,

$T > 16$ N

So, $F > 36$ N



For F = 30N,

For 5kg block,

$T = 30 - 20 = 10N$

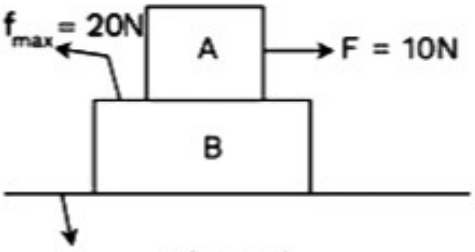
For 8kg block,

T = f. So, f = 10N → frictional force on 8kg block.

Q. **For the given system of blocks, find acceleration of 5kg block, 10kg block, friction between blocks and friction force between ground and lower block.**



Sol.



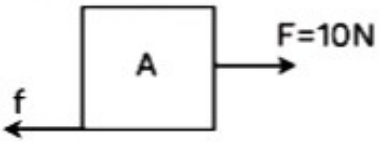
$$f'_{max} = \mu'(5+10)g$$
$$= 0.2 \times 15 \times 10 = 30N$$

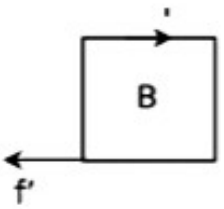
External force F = 10N is less than 20N.

Also, for the system to move, net force must be greater than 30N.

But 10N < 30N, so blocks cannot move.

$\therefore$ $a_A = 0, a_B = 0$





For equilibrium of blocks,

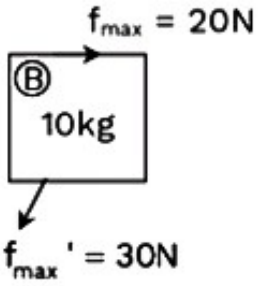
$f = 10N$

$f' = f = 10N$

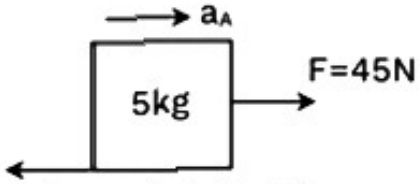
Q. **Find the acceleration of masses and frictional forces involved.**

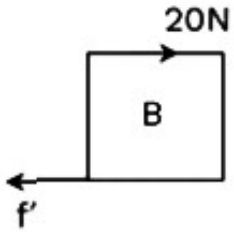


Sol.



As $f_{max} < f_{max}'$, so the 10kg block can never move.





So, $a_B = 0$

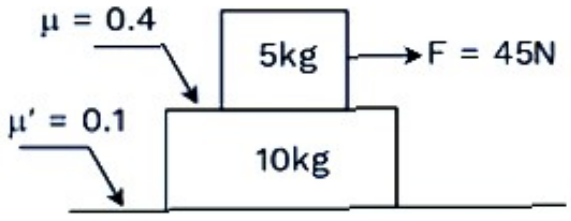
For block A,

So, $a_A = \frac{45-20}{5} = \frac{25}{5} = 5ms^{-2}$

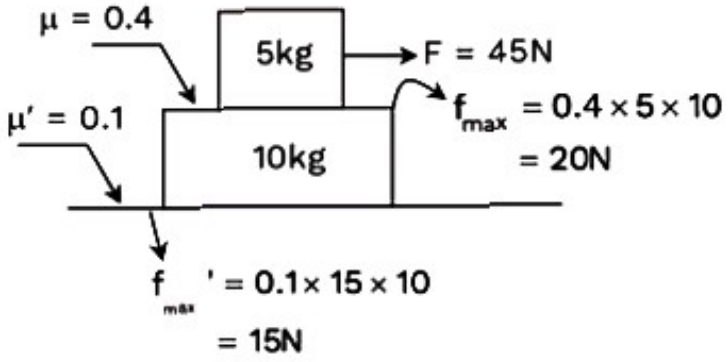
Friction force between blocks = 20N (Kinetic friction)

Friction force between ground and 10kg block = 20N (static friction)

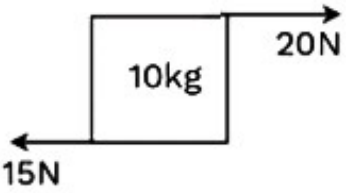
Q. **Find the acceleration of masses and frictional forces involved.**



Sol. Let's assume both blocks are moving together.



Then, $a = \frac{45-15}{5+10} = 2ms^2$

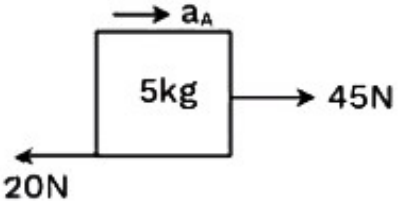


For 10kg block,

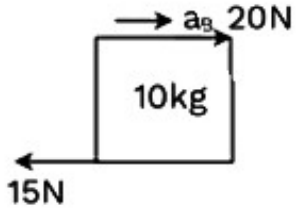
$a_{max} = \frac{20-15}{10} = 0.5m/s^2$

As $a_{max} < a$, so our assumption is wrong and both blocks cannot move together.

So, both the blocks are moving separately, and kinetic friction will act.



$a_A = \frac{45-20}{5} = 5m/s^2$



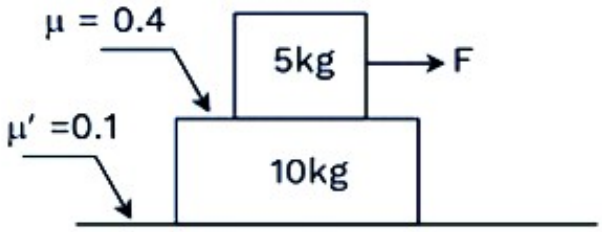
$a_B = \frac{20-15}{10} = 0.5m/s^2$

The frictional forces acting between the blocks are

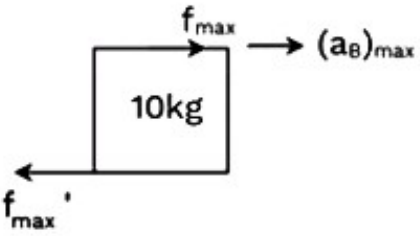
$f_{5-10} = 20$ N (kinetic)

$f_{10-G} = 15$ N (kinetic)

Q. **Find maximum value of F so that both blocks move together.**



Sol. For the 10kg block,



$$(a_B)_{max} = \frac{f_{max} - f_{max}'}{10}$$

$$= \frac{(0.4 \times 5 \times 10) - (0.1 \times 15 \times 10)}{10}$$

$$= 0.5 \text{ m/s}^2$$

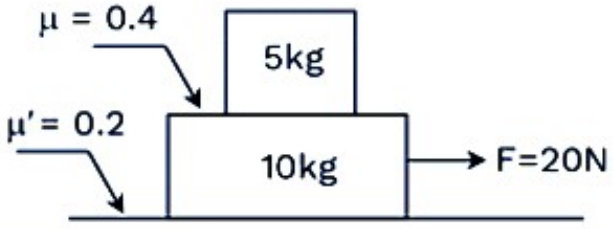
For both blocks to move together, maximum value of acceleration can be 0.5 m/s^2

So, 15 (0.5) = F – (0.1 × 15 × 10)

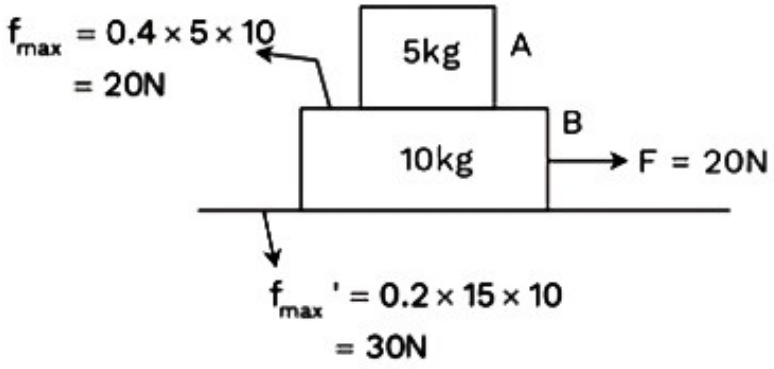
$$\frac{15}{2} = F - 15$$

$$F = 15 + \frac{15}{2} = \frac{45}{2} N$$

Q. **Find the acceleration of mass and respective values of friction forces.**



Sol.



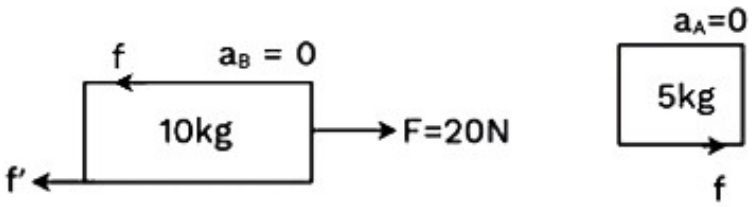
To break the friction on ground, 30N force is needed.

As F = 20N < 30N, so lower block cannot move.

Then, upper block will not move as well.

So, $a_B = 0$, $a_A = 0$

Now,

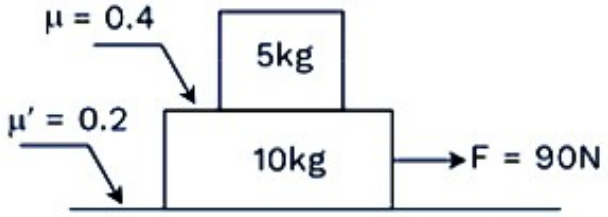


For upper block, $f = 5a_A = 5(0) = 0$

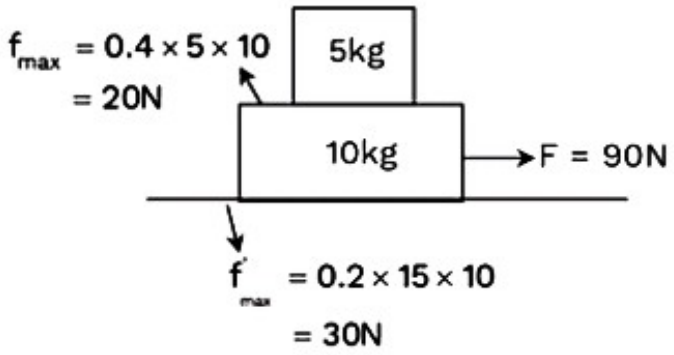
For lower block, $F - f - f' = 10a_B$

$20 - 0 - f' = 0 \Rightarrow f' = 20N$

Q. **Find the acceleration of the masses and respective frictional forces.**



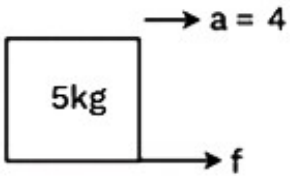
Sol.



Let's assume both the blocks are moving together.

Then, $a = \frac{90 - 30}{15} = 4 m/s^2$

For upper block to move with a = 4 m/s^2



$f = 5 \times 4 = 20$ N

Also, $f_{max} = 20N$

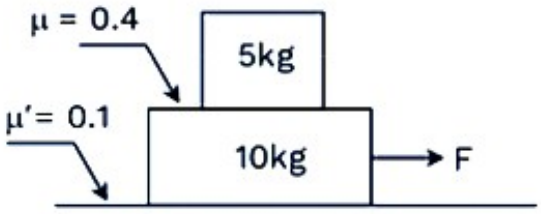
As $f = f_{max} = 20N$, so our assumption is correct and both blocks are moving together.

$\therefore$ $a_A = a_B = 4 ms^{-2}$

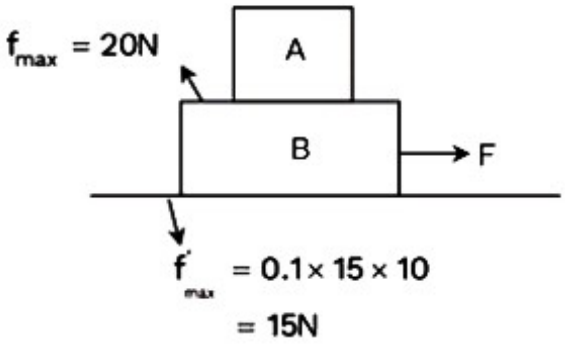
Friction force between the blocks, $f_{5-10} = 20N$

Friction force between lower block and ground, $f_{10-G} = 30N$

Q. **If acceleration of 10kg is 9 ms^{-2}, find the value of F.**



Sol.



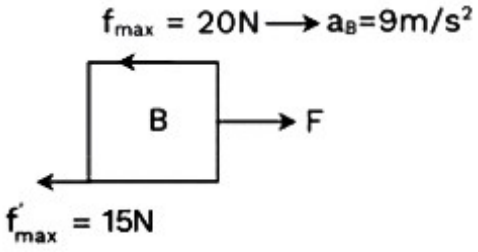
Maximum possible value of acceleration of A is,

$$a_{max} = \frac{20}{5}$$

$$a_{max} = 4\,m/s^2$$

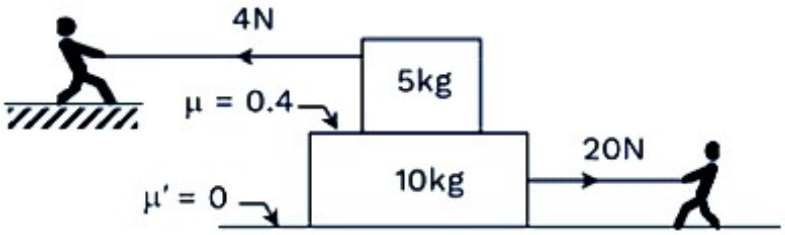
So, f_{max} is acting on A.

∴



F – 20 – 15 = 10 × 9
F = 125N

Q. **What will happen in the following system?**

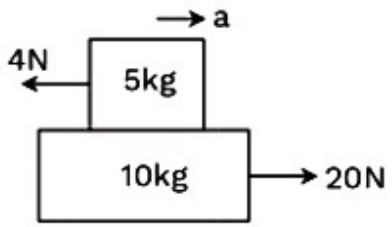


Sol. Assuming both blocks are moving together,

Then, $a = \frac{20-4}{15} = \frac{16}{15} m/s^2$

For the 5kg block,

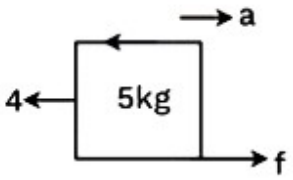
$$f_{max} = 0.4 \times 5 \times 10$$
$$= 20N$$



Now, for the 5kg block to move with $a = \frac{16}{15} m/s^2$,

$f - 4 = 5a$

$f = 4 + \frac{16}{3} = \frac{28}{3} = 9.33N$



As $f < f_{max}$, so our assumption is correct.

So, both blocks move together with common acceleration $a = \frac{16}{15} m/s^2$ and friction between them is $\frac{28}{3} N$.